

ESA ANNUAL REPORT 2021



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ESA Annual Report 2021

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Foreword

The ESA Annual Report is published yearly, as stipulated by the ESA Convention. Compiling a foreword to an Annual Report as Chair of the ESA Council is a suitable occasion to look back and take stock of what has been achieved. 2021 has been another extraordinarily successful year for ESA, once more characterised by excellence, partnership, and continuity.

ESA's excellence manifests itself across its many areas of scientific and technological expertise. Some of the highlights can be found in this Report. They embody the very essence of ESA, and its reason for existing — world class skills and leading-edge competencies, in established domains as well as in emerging fields. Proficiency in running complex space projects is a precious asset that ESA can bring to the table in an evolving framework of European space governance.

In this context, partnership is another key notion. ESA itself is a partnership between various countries. ESA's partnership with the European Union has been renewed and highlighted through the conclusion of the Financial Framework Partnership Agreement. Partnership is also the reason for the success of European flagship programmes like Galileo and Copernicus. And that partnership extends to trusting collaboration on a global scale, evidenced among other things through ESA's contribution to the launch of the James Webb Space Telescope aboard an Ariane 5.

Finally, continuity in ESA's role as a trusted partner is essential to securing a successful future. ESA is embarking on new routes, sticking to its mandate and leveraging its experience. At the same time, it cherishes its traditional competencies, thereby ensuring continuity. Sometimes change can bring continuity: by adapting its work processes in the context of the COVID-19 pandemic, ESA has made business continuity possible. With the new Director General Josef Aschbacher in post, ESA is moving forward — with continuity, stability, a new vision and new ideas.

Anna Rathsmann
Chair of the ESA Council

Director General Review

Looking back over 2021 obviously features a personal element for me, as it is the year when I became Director General of ESA. I was privileged to take over a strong Agency in excellent shape, thanks to its uniquely skilled and motivated work force and the continued, strategic support given by Member States.

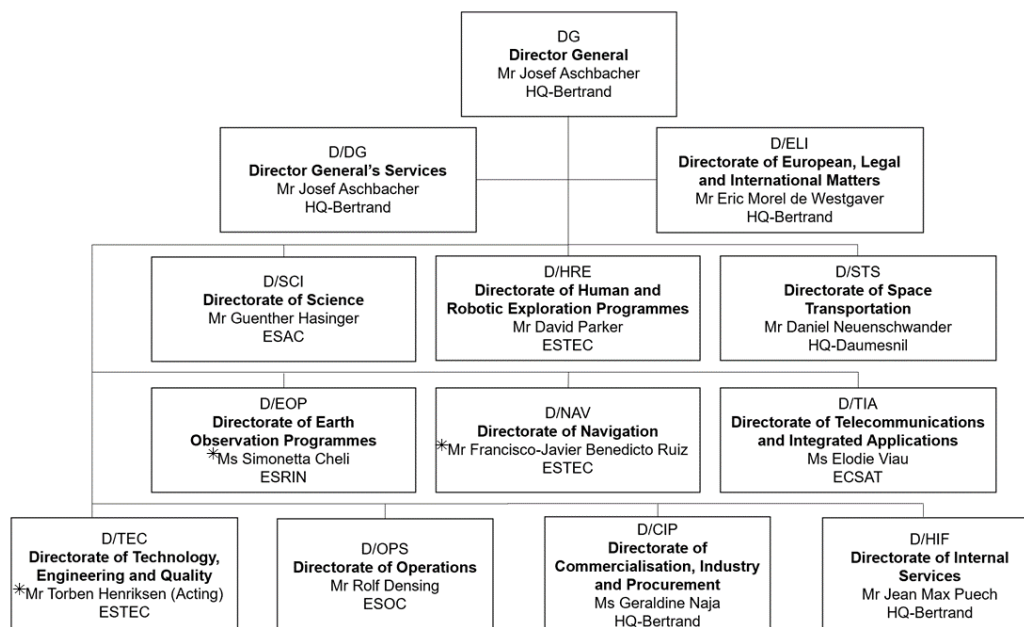
ESA has all the ingredients in place to continue its success story. At the same time the space sector is undergoing dramatic changes, often referred to as the New Space transformation: there are new actors, often from outside the traditional space sector; novel technical concepts such as mini satellites, constellations, or the use of artificial intelligence; easier access to space through reduced launch cost; and new ways of financing space projects, for example through venture capital.

In order to account for these challenges, we launched ESA Agenda 2025. It aims to unite the key European actors in space, beginning with ESA and the European Union. It also sets out to engage the private sector, and to facilitate access to funding. In addition, it undertakes to speed up processes and transform ESA so that it is well prepared for the future. ESA Agenda 2025 and the associated concepts have received the full support of the Member States, as expressed in the Matosinhos Manifesto.

So the strategic framework has been defined and a clear course of action has been set. But high-level orientations are pointless if there is no substance to underpin them. And that substance is provided by ESA on a daily basis — as clearly seen in 2021. The following pages bear notable testimony to this fact. ESA's 2021 Annual Report once more showcases the impressive expertise and unparalleled capacities of the Agency. They combine to make ESA a central, indispensable element of the European and worldwide space sector.

Josef Aschbacher
Director General

Management Structure



Organigram 2022, published on the 7th of February 2022.

* Directors not on duty at the end of 2021.

ESA Agenda 2025 Implementation

The Director General, on taking up duty on 1 March 2021, presented to the Member States “ESA Agenda 2025”, which was officially published at the beginning of April, defining five priorities (https://www.esa.int/About_Us/ESA_Publications/Agenda_2025). It describes the new Director General’s strategic vision for ESA until 2035. It is a plan for Europe to maintain and expand its leadership in space, addressing both current and future challenges. Calling for a common European vision as a matter of urgency, ESA Agenda 2025 underlines that Europe must aspire to space activities and programmes that are absolutely world class. In this context, Europe is understood to comprise all relevant actors: ESA Member States, the European Commission, other European institutions, space and non-space industry, the science and research communities — and all the people involved. Sixty actions were defined in order to measure progress in the implementation of ESA Agenda 2025. Some progress has already been achieved in 2021 on those five priorities.

Strengthening ESA-EU relations:

The Financial Framework Partnership Agreement (FFPA) between the European Union and ESA was finalised and signed in July 2021, clarifying the respective roles of ESA and the EU. A High-Level Advisory Group was put in place and produced a report which urged ESA to accelerate the use of space in Europe while identifying three “Accelerators” and two “Inspirators” (https://esamultimedia.esa.int/docs/corporate/Accelerating_the_use_of_space_in_Europe.pdf). An Intermediate Ministerial Meeting was organised on 19 November in Matosinhos (Portugal), resulting

in the unanimous adoption of the “Matosinhos Manifesto” (<https://vision.esa.int/the-matosinhos-manifesto-accelerating-the-use-of-space-in-europe/>) and providing ESA with a mandate to further study those Accelerators (Space for a Green Future, Rapid and Resilient Crisis Response, Protecting our Space Assets) and Inspirators (Icy moon sample return mission and European human space exploration).

Boosting green and digital commercialisation

A certain number of actions have been defined to support the emergence of a more vibrant and dynamic commercial space ecosystem to successfully bring about the green and digital transformations. In particular, a new Commercialisation Department has been created that will define an ESA-wide approach through a range of actions (supporting market creation efforts for new applications; strengthening ESA’s role as first buyer or anchor customer on behalf of its Member States and the EU for new commercial services; providing industry with a procurement forecast for space products and services; proposing concrete actions to offer ESA expertise as technical partner/adviser in new cooperative schemes with funding entities such as venture capital funds and business angels; opening up access to ESA’s unique technical facilities; simplifying access to ESA programmes and activities for start-ups, SMEs, mid-caps, and New Space companies; reducing significantly the average time-to-contract for contracts below €1m; increasing the use of prizes and grants; updating the ESA Technology Strategy to serve commercialisation and innovation targets; creating a dynamic network of ϕ -labs across ESA and across Europe).

Developing space for safety and security

The pillar that was defined for the 2019 Council meeting at ministerial level is being reinforced in the framework of the preparation for the 2022 Council meeting at ministerial level (CM22), with activities complementing efforts at national and EU level. Programme proposals are being drawn up in the areas of space traffic management (contributing to the collision risk estimation and mitigation programme), debris mitigation and removal, space weather, planetary defence and space logistics.

Specific efforts were made in 2021 to further develop and upgrade active cyber technologies, ensuring the availability, integrity and confidentiality of space-based systems and data.

Actions also got under way on the reinforcement of activities supporting security on Earth with the definition of several programme proposals. A conference on space for security on Earth involving national security stakeholders was organised on 4–5 November 2021 to obtain from “users” an assessment of their needs.

Addressing critical programme challenges

A certain number of challenges have been identified in all domains, some of which are addressed in the daily activities and reported in the relevant chapters, with several included in the draft programme proposals presented to the Member States at the end of 2021, in view of the discussion preparing for the subscriptions expected at CM22 in November 2022.

Completing the ESA transformation

In order to be able to implement all actions defined for the first four priorities, ESA has to evolve and to transform itself. Some of these transformation actions have already begun seeing first results in 2021, at least in terms of the analysis of what has to be done. Areas covered include data management, improving financial management and processes, digitalising project management,

corporate social responsibility and the reduction of greenhouse gas emissions, improving gender balance and inclusiveness, creating a Junior Professional Programme and being inspirational for younger generations.

A dashboard has been developed showing the status of each of the 60 actions, accessible to the ESA workforce and to Member State delegates and was made available in mid-December. It will allow a follow-up of the progress of those actions.

Programmes and Activities

Science & Exploration

Science

Throughout 2021, the Science Programme continued to provide the European scientific community with the tools to achieve and maintain excellence in a number of key fields. The Gaia mission continued to scan the sky to create the most precise 3-dimensional map of the Galaxy, and while only an interim version of its data products has as yet been released, the data from Gaia have resulted in an unprecedented 1300 scientific publications appearing in 2021. While the accuracy of the interim data products is still limited with respect to the final data products, they have already enabled investigations that could only be imagined until now, such as the direct measurement of the acceleration of our Sun's motion around the centre of the Milky Way (a vanishingly small value, equivalent to 2% of one billionth of the gravitational acceleration at the Earth's surface). One can only imagine what the scientific productivity from the final Gaia data products will be!

New, ambitious missions that will provide the European scientific community with new tools to achieve and maintain excellence in their fields are nearing launch, with both the JUICE and Euclid missions being prepared for launch in early 2023 (with a slight delay from the previous plan for launch in 2022). JUICE will be the first mission to enter into orbit and study from close up the moon of another planet: Ganymede, one of the moons of Jupiter originally discovered by Galileo, has an icy surface covering an ocean of liquid water, and is thus a possible site hosting life in our Solar System. JUICE, building on the previous European successes in the exploration of the remotest corners of our Solar System, for example the Rosetta mission to a comet, will carry on board a suite of sophisticated instruments to focus on this fascinating and remote icy moon.

Euclid will instead map the sky to build a 3-dimensional map of distant galaxies that will be used to investigate the nature of the elusive "dark energy" that is causing our Universe to expand at increasing speed, countering the effect of gravity that would naturally cause its expansion to slow down. Dark energy has been observed to be the dominant "ingredient" of the Universe, yet its nature remains obscure. More generally, Euclid's high-resolution map of the deep Universe will provide a unique tool to study the evolution of a large variety of objects in our Universe and will serve as a reference for the future.

Two other missions, both dedicated to observing exoplanets, have progressed significantly, with the PLATO "planet searcher" (due for launch in 2027) approaching its critical design review, and the prime contractor having been selected for the Ariel "planet atmospheric analyser" (due for launch in 2029). Both missions will give European astronomers leadership in their respective areas of exoplanetary research, and will complement and consolidate European participation in the James Webb Space Telescope in this area, together with the work being carried out by the small mission CHEOPS.

Looking further into the future, a new planetary mission, EnVision, has been selected in the Science Programme. EnVision, to be developed in partnership with NASA, will study the elusive, cloud-shrouded surface of Venus using a high-resolution synthetic aperture radar (SAR).

The longer-term future of the Science Programme was laid out in 2021, with the scientific community setting out their priorities in the “Voyage 2050” plan approved by the Science Programme Committee (SPC) in June, for the two decades following the launch of the Athena and LISA flagships. The plan, the result of almost two years of work from a broad segment of the European space science community, defines, among other things, the scientific themes for the three future flagship missions (“Moons of the Giant Planets”, “From Temperate Exoplanets to the Milky Way”, and “New Physical Probes of the Early Universe”), each staking a claim for European leadership in a key area of space science.

2021 finished for the Science Programme on a brilliant note, with the picture-perfect launch on board an Ariane 5 from Kourou of the James Webb Space Telescope, the biggest space telescope ever built. Webb, a NASA-led mission, carries two European-led instruments, the NIRSpec and the MIRI spectrographs, the two instruments that will be able to analyse the light from the sources that Webb will observe, from the most remote and oldest galaxies in our Universe to nearby exoplanets. Thanks to the partnership with NASA and the CSA, European astronomers will have guaranteed access to a unique facility that will remain unequalled for several decades to come.

Human Spaceflight and Robotic Exploration

Highlights

The European Columbus Ka-band (CoKa) terminal was installed outside the Columbus laboratory during an extra-vehicular activity (EVA) on 27 January. CoKa will provide ESA with independent high-speed data relay via the EDRS (European Data Relay System) network.

NASA’s Perseverance rover landed on Mars on 18 February marking the start of the joint NASA-ESA Mars Sample Return (MSR) campaign which will return to Earth Martian rock and regolith samples for the first time. Video data of the landing was relayed by the ExoMars Trace Gas Orbiter (TGO) spacecraft. Communication between TGO and Perseverance takes place typically twice a day.

The Astronaut Selection Campaign was announced by the ESA Director General on 16 February and the corresponding job requisition was published on 31 March with a closing date of 26 May. More than 23,000 applications were received including over 200 for the parastronaut feasibility project. The first round of testing for the 1,391 candidates who made it through the initial screening focused on cognitive, technical, motor coordination and personality tests. For successful candidates, this will be followed by a set of psychological interviews and group tests before medical testing.

Education activities engaged tens of thousands of school students across Europe and trained thousands of teachers to use space in the school context.

1 Cornerstone 1: Humans in LEO

The 11m-long [European Robotic Arm](#) (ERA) was launched on a Proton rocket from Baikonur on 21 July attached to the Multipurpose Laboratory Module (MLM), “Nauka”, and was docked manually by the crew on 29 July.

Thomas Pesquet was launched from Kennedy Space Center (KSC) in Florida on the SpaceX Crew-2 Dragon atop a Falcon 9 rocket on 23 April. During his six-month “Alpha” mission he performed several EVAs and assisted in numerous experiments. From 4 October he was commander of the ISS until his return to Earth on 9 November.

Matthias Maurer and the members of Crew-3 were launched from KSC aboard the SpaceX Crew Dragon spacecraft “Endurance” on 12 November. Matthias named his six-month mission “Cosmic Kiss”. Samantha Cristoforetti continued training for her “Minerva” mission, planned for launch in April 2022.

Technology demonstrations on the ISS continue including the Analysing Interferometer for Ambient Air 2 (ANITA-2) which was launched on SpX-24 on 21 December and installed in the NASA Express Rack.

2 Cornerstone 2: Humans Beyond LEO

Multi-Purpose Crew Vehicle European Service Module 1 (MPCV-ESM-1) was transferred to the Vehicle Assembly Bay in KSC where it was integrated with the Space Launch System (SLS) and tested. The Artemis1 launch is expected to take place in April 2022.

ESM-2 was delivered from ADS Bremen to The Kennedy Space Center in Florida in October. ESM-2 will provide propulsion, power and thermal control for the Artemis-3 mission while supplying the astronauts with water and oxygen on their flight around the Moon. Launch is planned two years after Artemis 1.

The procurement of recurring equipment and subsystems for ESM-3 is in progress. ESM-3 will support Artemis 3, the planned demonstration human lunar landing. The European contributions to the NASA Gateway (I-HAB and ESPRIT) have entered development. A larger European Refuelling Module (ERM XL) was defined.

3 Cornerstone 3: Moon Robotic Exploration

The PILOT-D landing camera demonstrator on board Luna-25 was successfully switched on in December. The Roscosmos lander launch is planned for July 2022.

International Memorandums of Understanding for the provision of European payloads on NASA Commercial Lunar Payload Services missions and for the JAXA/ISRO LUPEX mission were approved by Council in December.

4 Cornerstone 4: Robotic Mars Exploration

ExoMars 2016 is in flight operations. The Trace Gas Orbiter (TGO) spotted significant amounts of water at the heart of Mars’ canyon system, [Valles Marineris](#). The water, which is hidden beneath Mars’ surface, was found by [TGO’s Fine Resolution Epithermal Neutron Detector \(FREND\) instrument](#), which is mapping the hydrogen in the uppermost metre of Martian soil.

ExoMars 2022 development progressed towards launch in September 2022. The Rosalind Franklin rover completed all tests in preparation for shipment to Baikonur. Two successful high-altitude parachute drop tests took place in Oregon (US) in November and December.

Intensive work continued throughout 2021 on the European contributions to **Mars Sample Return (MSR)**: the Earth Return Orbiter (ERO), Sample Fetch Rover (SFR) and the Sample Transfer Arm (STA).

5 Science in Space Environment (SciSpace)

Dozens of experiments were performed on the ISS in 2021 while science campaigns using the drop tower, parabolic flights and other ground-based facilities continued. The programme involves nearly 2,000 European scientists.

6 Exploration Preparation, Research and Technology (ExPeRT)

Preparation of the EL3 Moon lander project continued and key technology work was started. In December, three Requests For Information (RFIs) were released:

- Services for Human Space Exploration – Meeting ESA’s future Research and Astronaut Mission Objectives in Low Earth Orbit and Beyond (jointly issued by the HRE and STS Directorates)
- Future European Missions and Elements of Lunar Exploration
- Future Elements of Mars Exploration

Applications

Earth Observation

Earth Observation and COVID-19

EOP activities were carried out nominally throughout 2021. Interactions and meetings with partners continued as in pre-COVID times, albeit mostly virtual or hybrid.

The Rapid Action on coronavirus and EO (RACE) dashboards on Earth Observation (EO) and COVID-19 impacts kept growing and were being updated with current information. The tri-agency dashboard set up by ESA, NASA and JAXA featured numerous economic, agricultural and water-related indicators.

Copernicus

Copernicus Sentinel missions continued to provide valuable data and information on the state and health of the planet. More than 507,000 users have registered on the open access hub and 413 PB of data had been downloaded by the end of 2021. These numbers keep growing, and only represent the tip of the iceberg, as many data are used directly by the Copernicus services or mirrored on other sites.

Sentinel-1A has exceeded its design lifetime of 7 years of operations and will continue its operations until Sentinel-1C is fully operational. Copernicus Sentinel-1B has been unavailable since 23 December 2021. Investigations into the anomaly are ongoing, to determine the root cause of the loss of power and identify possible solutions. After a successful commission phase, Sentinel-6 Michael Freilich products became available in December.

Sentinel development activities including Copernicus Expansion mission Phase B activities have progressed nominally. On 22 June, finalisation of the Financial Framework Partnership Agreement / Contribution Agreement was celebrated in a special event in Brussels. For the Copernicus Expansion missions under preparation, industrial activities at both prime and subcontractor level progressed nominally in 2021 with Preliminary Design Reviews for all the Sentinel Expansion Missions expected in 2022.

The amount of the EU’s contribution fell short of contributions from non-EU ESA Member States and ESA has supported the European Commission in the analysis of options. ESA and the Commission planned to propose a way forward at the 6th EU/ESA joint PB-EO/Copernicus Committee seminar meeting on 20 January 2022, which, if welcomed by Member States, would form the basis of a DG proposal on the Copernicus Space Component way forward and subsequent amendment to the Contribution Agreement between the Commission and ESA.

Earth Explorers / Science / Other Missions

The Earth Explorers in orbit continued to deliver crucial data for science, within the remit of their original scope and beyond. On 29 November, the Aeolus satellite and its ALADIN payload reached the formal completion of its nominal mission lifetime. On 22 November, the three Swarm satellites celebrated their eighth year in space. On 14–17 June 2021, 10 years of CryoSat was celebrated with a science conference. The process to extend the SMOS, CryoSat and Swarm missions into Segment 2, the next programmatic period of FutureEO, has begun.

Preparation of future missions has continued as planned. Phase A of the Earth Explorer 10 candidate mission, Harmony, is ongoing with two parallel system studies, science support activities and campaigns. The Aeolus Follow-On Instrument Consolidation Study has taken into account the user requirements being consolidated with EUMETSAT, the lessons learned from Aeolus development and in-orbit operations and the accommodation compatibility for both lasers being developed in the parallel predevelopments. Significant progress has been achieved in the preparation of the Next Generation Gravity Mission (NGGM) as a Mission of Opportunity in cooperation with NASA, also in the framework of the Joint Programme Planning Group, to enable a Mass change and Geosciences International Constellation as studied with NASA and the user community since the last decade.

Progress on the selected Earth Explorers under development has been nominal, allowing for launch timeframes of Biomass and EarthCARE in late 2023, FLEX in mid-2025, and FORUM at the end of 2027.

A contract was signed with Arianespace to secure the Vega-C joint launch in mid-2025 for FLEX and Atmospheric Limb Tracker for Investigation of the Upcoming Stratosphere (ALTIUS).

The implementation phases for Scout-2 (HydroGNSS) and for Scout-1 (CubeMAP) were successfully initiated (planned launch dates in 2024).

TRUTHS Phase B1 system studies proceeded at full speed making it possible to initiate the process for a Gate Review in July 2022.

The FSSCat mission, launched in 2020, completed the in-orbit commissioning of FSSCat/ Φ -sat-1, including successful in-flight demonstration of accurate cloud detection with an AI chip.

As planned, after eight years of successful operations, the Vegetation instrument data operations on Proba-V stopped on 31 October, except for very limited and specific data acquisitions.

The Global Monitoring of Essential Climate Variables Climate Change Initiative (GMECV/CCI) contributed significantly to the Intergovernmental Panel on Climate Change (IPCC) Assessment Report (AR6) on The Physical Science Basis released in August.

A revision of the Global Development Assistance (GDA) Implementing Rules were approved by Council in October and the first contracts on Disaster Resilience and Climate Resilience have been kicked off.

For the InCubed Element (InCubed & InCubed+ together), there are 78 activities in the overall pipeline, with 40 activities running and four having been completed (three of which have generated commercial revenues).

Meteorological Missions

For MTG-I1, all efforts have continued to ensure a start of the Qualification Acceptance Review in June 2022 with associated launch before the end of 2022 so as to be able to secure an Ariane 5 launch.

As for MetOp-SG, all Satellite-A PFM Platform units have been integrated (with the exception of the Solar Array) as well as the Radio Occultation Prototype Flight Model, METImage fEEM and Sentinel-5 IIM instruments.

All critical suppliers for the Arctic Weather Satellite (AWS) payload have been kicked off and work progressed leading to a start of the Preliminary Design Review at the end of 2021.

Transversal / Partnerships / International Cooperation

ESA Phi-Week was held from 11–15 October. The event was held in hybrid format and gathered over 2,300 online attendees from 84 countries.

At the conclusion of the 2021 CEOS plenary meeting (1–4 November 2021), ESA assumed the role of CEOS SIT Chair for a period of two years (2022–2023), succeeding Australia.

The Earthnet Third Party Mission (TPM) activity continued to be highly appreciated by scientists and applications developers with more than 10,000 research projects having used TPM data since 2008 and almost 1,000 new TPM users registering in the last 12 months. Earthnet also successfully supported the International Charter Space and Major Disasters.

30 years since the launch of ERS-1 resulting in a long series of ERS Heritage data available to users was highlighted in several articles, video and a new ERS infographic, all published on ESA websites in July.

ESA was well represented at COP26 in Glasgow in November, through several side events, interviews and a presence in the blue and green zones.

The European Commission and the ESA signed the Contribution Agreement for the implementation of the Destination Earth (DestinE) initiative.

Preparations took place for the Living Planet Symposium (LPS) in Bonn, Germany, to be held on 23–27 May 2022 in Bonn, Germany.

Simonetta Cheli was selected as Director of Earth Observation and Head of ESRIN in Frascati, IT, as of 1 January 2022.

2021 saw continued consultation with Member States to prepare Programme Proposals for CM22. Also, a dedicated workshop on EO Programmes was organised on 6 December.

Space Safety & Security

Space Safety

2021 was a very important and busy year for Space Safety, which made its way to the top of the political agenda and became the fastest growing programme in ESA.

Space Safety had a prominent role at the Intermediate Ministerial Meeting held in Matosinhos, Portugal, where Ministers unanimously adopted a Resolution to accelerate the use of space in Europe (the “Matosinhos Manifesto”). The Resolution defined three “Accelerators” the third of which, called “Protection of Space Assets” (PROTECT) will have the objective of safeguarding European assets in space and on ground from interference caused by space weather and space objects.

Space Safety was also part of the new Financial Framework Partnership Agreement (FFPA) signed by ESA and the EU in June. A collaboration on Space Weather and Planetary Defence topics was agreed whereby a number of important tasks from the European Union Space programme and Horizon Europe were entrusted to ESA.

ESA Agenda 2025, published in April, included Space Safety as one of its five top priorities, with clearly defined objectives including the development of new standards, the contribution to a global space traffic management system, the development of technologies for in-orbit servicing, and a strong push for commercialisation.

Space Weather and Lagrange

The two space weather payloads SOSMAG, installed on the Korean GEO-KOMPSAT-2A satellite as part of the Korean Space Environment Monitor, and NGRM, hosted on the second satellite of the European Data Relay System (EDRS) constellation, began their operations and data was made available to the community.

A number of units (radiation monitor) were under integration in the host spacecraft or under preparation for new missions (Lunar Gateway).

The new contract for the Space Weather Service Network with over 50 participating entities was signed in July and a major update of the Space Weather Service Portal was released in October.

A Space Weather End User Forum was held with 50 end users actively participating, with a range of backgrounds including aviation, spacecraft operation, human spaceflight, insurance, civil protection and media/communications.

Concerning the Lagrange mission, the proposal for Phase B2, C/D and E1 was received from industry in October and evaluation began in combination with the system requirements review. The Lagrange Mission Advisory Group (MAG) was established and the first meeting took place in July.

Planetary Defence and Hera

The NEO Coordination Centre (NEOCC) was inaugurated at ESRIN. ESA's second Test-Bed Telescope, hosted at the La Silla Observatory in Chile, saw its "first light".

A number of near-Earth objects were discovered in 2021 using ESA's telescope network, including the asteroid with the shortest orbital period around the Sun. There was also an international exercise to observe asteroid Apophis, with a close flyby of 30,000 km above the Earth.

Several international exercises simulating asteroid impact threat scenarios and the (fake) discovery of an asteroid which would require a deflection activity were carried out to test the interfaces and cooperation of international entities in response to these situations.

Concerning the Hera mission, NASA's companion DART mission was successfully launched on 24 November, and the first acquisition performed by ESA's Estrack network, with ESA up-linking the first telecommand to the DART spacecraft. The preliminary design review of the Hera spacecraft was successful.

Space Debris, Clean Space, Collision Risk Estimation and Automated Mitigation (CREAM) and Active Debris Removal and In-Orbit Servicing (ADRIOS)

In 2021, a Russian missile test generated 1,500 new debris fragments and a Chinese rocket's upper stage made an uncontrolled re-entry over the ocean. Also, the continuing increase of close approaches between ESA satellites and other operational satellites during the summer was affected by missing information and delayed availability of the other operators. These events highlighted the importance of developing space debris and space traffic management (STM) technologies and international regulatory frameworks.

A statement of intent on potential cooperation in conducting space-based observations of small-sized space debris was signed by ESA and NASA in August.

Member States of EU and ESA met at the STM Conference in July, where they agreed on fostering a common European approach to STM.

The contract for the debris removal mission ADRIOS/Clearspace-1, was signed in early 2021, and work proceeded according to plan, achieving IMM1 (Intermediate Milestone). The mission showcases a new approach to ESA procurement in which the removal of an ESA-owned object is procured from industry as a turn-key service.

The call for proposals for a new in-orbit servicing (IOS) mission was issued at the end of September, with the requirement of having established relations with an identified client. The selected mission will be part of the Space Safety Programme proposal for CM22.

Security and Cyber Resilience

In an ever-evolving global security context, space infrastructure has become critical to the everyday lives of European citizens and society. ESA is strengthening its capability to integrate this new security context and guarantee full reliability in developing secure, robust space assets, missions and technologies and the availability, integrity and confidentiality of the data and services these provide. Thus, the interests and investments of our Member States, and society as a whole are safeguarded. The ESA Security Office (ESO), via the ESA Security Framework, has centralised security within a crosscutting multi-directorate coordination, ensuring it forms the basis of all ESA activities and ensures a correct ESA security posture.

Cyber Resilience

Space applications and services have become political drivers and key assets for the economy, security and the daily lives of European citizens, rendering space infrastructure critical and exposed to deliberate malicious attack. ESA's Cyber Security Strategy aims to provide adequate protection of the space domain, establishing a unique cyber capability in Europe composed of:

- ✓ **Security Cyber Centre of Excellence (SCCoE)**
- ✓ **Cyber Security Operations Centre (C-SOC)**
- ✓ **Cyber Security Coordination** – a multi-domain, cross-directorate roadmap of cyber security activities has been established up to 2027.

The European Space Operations Centre (ESOC) Mission Operations Infrastructure (MOI) covers all domains of the ESA Flight Operations Ground Segments. In order to ensure the ability of ESA's MOI to continuously deliver highly reliable services, despite adverse cyber events in a global climate of steeply increased risk of cyber attacks, a number of measures have been put in place, such as:

- The ESA Operations Directorate continued to successfully maintain its ISO2700-compliant MOI Information Security Management System (ISMS), necessary to ensure protection of the confidentiality, availability, and integrity of assets against threats and exploitation of vulnerabilities. A successful recertification was achieved in 2021.
- The MOI network security policy for legacy access to operational and preoperational networks was ensured.
- Several mitigation measures were successfully implemented in response to critical security vulnerabilities discovered worldwide.

Furthermore, the ESA Operations Directorate (OPS) significantly contributed to the Threat and Risk Assessment for upcoming missions such as the Earth Return Orbiter (ERO), next-generation Copernicus, and the Space Safety Programme (S2P) Lagrange mission to ensure full compliance with the ESA Security Framework. The aim of the assessment was to identify and qualify cyber security threats to the missions and to develop a clear set of countermeasures required to establish an adequate level of cyber resilience.

In terms of cyber resilience innovation, OPS has initiated a novel process to diversify cyber resilience through moving towards a defence-in-depth concept. The approach recognises each system layer, from networks to software, as playing a role in the overall cyber resilience posture.

OPS continues to be at the forefront of cyber security innovation by leading R&D in the fields of secure systems engineering and experimental next-generation secure space link protocols, where several In-Orbit-Testing (IOT) experiments have been successfully executed using the ESA OPS-SAT mission.

Space Programme Accreditation & Certification

ESO accredits and certifies ESA programmes, applying the ESA Security Framework, thereby supporting ESA programmes in providing adequately secured space systems (in terms of availability, integrity and confidentiality), developed in secured environments through a holistic, coordinated approach.

Corporate Security

Security risk management is applied across all ESA business such that security risk is accurately understood and appropriately mitigated to preserve ESA's security risk tolerance when carrying out ESA projects.

ESO has successfully performed the Annual ESA Security Inspection Plan, providing assurance for ESA sites and assets.

Telecommunications and Integrated Applications

Telecommunications and Integrated Applications

The ARTES 4.0 programme strengthens ESA's support to the competitiveness and commercialisation of the European satellite communication industry and the space industry at large. Four new partnership projects were initiated in 2021 and nine launches took place: SES17 with Neosat, Eutelsat Quantum, and seven nanosatellites with the Pioneer programme, five with Spire, one Open Cosmos and one with In-Space.

ARTES 4.0 Strategic Programme Lines

- ***Space for 5G***

A **5G/6G Hub** at the ESA ECSAT site in the United Kingdom connects 5G industrial communities and offers a collaborative space to tackle challenges and opportunities.

The **Sunrise project**, a partnership project with OneWeb (UK) underwent a successful design review. The first launch is planned for late 2022.

5G-EMERGE, a public-private partnership project for 5G broadcast, led by the European Broadcasting Union including 20 industrial partners, was initiated in 2021.

- ***Optical Communication-ScyLight***

ESA fosters European leadership through system concept, technology and product development to unleash the full capabilities of optical communication. ESA is building the world's first high-throughput optical network (HydRON) and Phase A/B1 was started in 2021.

- ***Space Systems for Safety and Security" (4S)***

Advanced satcom solutions will allow Europe's digital infrastructure to be ubiquitous, more secure and robust.

The Iris project addresses air traffic management. The end-to-end Iris system validation campaign is ongoing. Iris as Global is under final negotiation and will provide further geographical reach and service extension.

QKDSat is a partnership with ArQit Ltd aiming at providing commercial services for the distribution of highly secure encryption keys. In 2021 the System and User Segment preliminary design reviews (PDRs) were successfully concluded.

Activities with GovSatcom Precursor to contribute to European GOVSATCOM initiatives with Pacis 3 were pursued and the critical design review (CDR) successfully completed.

Positive steps were made towards EU-ESA cooperation to develop and validate a first-generation Secure Connectivity Constellation. A GOVSATCOM contribution agreement with the European Commission was also agreed.

ARTES 4.0 Generic Programme Lines

- ***Partnership Projects***

Under the **Novacom I – OneSat Partnership Project**, Airbus is developing and qualifying its OneSat platform product line, a standardised powerful software-defined satellite. The final design review was completed in 2021 and seven OneSats have been sold.

Under the **Novacom II – Space Inspire Partnership Project**, Thales Alenia Space is developing and validating innovative European subsystems and equipment for its software-defined satellite. Three Space Inspire satellites have been sold to date.

Quantum – Eutelsat Quantum is the world's first full software-defined satellite, preparing for the next generation of satellites reprogrammable in orbit. The satellite was launched on an Ariane 5 flight and commercial services started in November.

Triton-X – A commercial mission opportunity and a new microsatellites platform product line, called Triton-X, began its Phase C in 2021.

European Data Relay System (EDRS) – The EDRS satellites provide full commercial service for four Copernicus Sentinels, with over 57,000 optical intersatellite links performed to date. An EDRS-ColKa terminal on the International Space Station began services in December.

Next Generation Platform (Neosat) – The Neosat Partnership Project covers the development of **Spacebus Neo** for TAS, and **Eurostar Neo** for Airbus. Sixteen Neosat-based satellites have been sold so far and three have been launched, with a return exceeding €20 for every euro invested by Member States.

Electra – The objective of the Electra project is to develop an innovative “full electric propulsion” platform in the three-tonne launch mass range; the focus in 2021 was on the closure of the platform critical design review. Two leads are under consideration for a flight opportunity.

- ***Business Applications and Space Solutions (BASS)***

ESA Space Solutions exists to bring about the commercial exploitation of space assets. The Programme manages over 150 activities and 300 start-ups. Two ESA Investor Forums were held, raising over €65m from 35 investors and corporations. A Green Dossier with roughly 90 activities demonstrates how space data support the monitoring of environmental parameters and economically sustainable solutions.

A Space Solutions socio-economic impact analysis showed, with €368m of ESA Member States’ investment, including Business Incubation Centres (BICs) funding around 2,000 companies, helping to create over 5,900 direct jobs, generating €1.4 billion in revenues and securing about €970m of private investment.

- ***Core Competitiveness***

With more than 320 contracts under way, European industry has pioneered disruptive technologies such as AI, cloud, virtualisation, user terminal miniaturisation and flat self-scanning antennas. Core Competitiveness delivers a high return on investment, with a return on sales of 2.8 to date and 9.7 as a five-year projection. Some 1,700 jobs have been created and another 1,200 are expected to be created in the coming years. The programme worked with 80 SMEs in 2021.

Navigation

Implementation of FFPA

Transition between the GSA-ESA Galileo Working Arrangement and the EUSPA ESA GNSS Contribution Agreement has been completed pursuant to Article 49 and Annex XV of the FFPA (Financial Framework Partnership Agreement). By 1 November, all eight Specific Contracts under the GSA-ESA Galileo Working Arrangement had been transferred seamlessly into the FFPA Contribution Agreement or closed.

ESA staff members contributing to Joint Office tasks for Galileo and EGNOS as defined under Article 24 and Annex XIV of the FFPA have been nominated and took up duty on 1 October, in support of the project management tasks of the Commission.

ESA-EUSPA integrated teams as defined under Article 25 and Annex II of the FFPA have been set up in the context of EGNOS and Galileo, in support of ESA system development tasks and EUSPA exploitation tasks, respectively.

Galileo

Galileo satellites 27 and 28 were successfully launched on 5 December. Subsequently the LEOP (launch and early orbit phase) and in-orbit testing under EUSPA responsibility was supported by the combined ESA/OHB satellite team, which will lead to in-orbit test review of these satellites and eventual integration into the operational constellation in March 2022.

Procurement and testing of the remaining eight Galileo Batch 3 satellites is progressing according to plan towards Launch 16.

Galileo G1 activities progressed largely in line with agreed Full Operational Capability (FOC) Programme roadmaps, albeit with some delay due to late availability of site hosting services related to GSMCs (Galileo Security Monitoring Centres) and GCC (Galileo Control Centre).

Galileo G2 procurement is progressing in line with agreed Programme schedules. Notably the two G2 satellite development contracts have successfully passed their preliminary design review (PDR), and contracts related to procurement of G2 on-board atomic clocks have been kicked off. Procurement actions related to G2 ground segment and Test Beds were initiated in December, with step1 proposals due within weeks.

Galileo G1 and G2 industrial procurement actions initiated by ESA on behalf of the European Commission in 2021 amounted to €2.428 billion.

EGNOS

The GSA-ESA specific contract for EGNOS V3 development was formally terminated on 1 November in line with the FFPA transitional provisions. In the framework of the FFPA, ESA is in charge of the EGNOS V3 Industrial Contract in the name and on behalf of EUSPA.

During the second part of 2021, in agreement with EUSPA, ESA decided to implement specific measures in terms of technical support and co-engineering with Industry and/or EUSPA to further secure the overall end-to-end EGNOS V3 schedule.

EC/ESA H2020/Horizon Europe upstream navigation R&D

The H2020 Delegation Agreement is still in force up to the end of 2025, with the ability to award contracts extended up to December 2022.

The first Horizon Europe workplan to be funded by the new European Commission framework, set up for 2021–2028 since the FFPA signature in June, was submitted on 11 Oct and is currently awaiting approval by the European Commission.

Navigation Innovation and Support Programme (NAVISP) and preparation of CM22

In 2021, NAVISP Phase 1 and Phase 2 were implemented according to plan.

Building on the success of the NAVISP programme, the continuation of NAVISP into Phase 3 has received very wide support within PB-NAV (Programme Board on Satellite Navigation) in terms of content and budget envelope.

Enabling & Support

Space Transportation

Exploitation (Ariane 5, Vega and Soyuz)

Three Ariane 5, three Vega and one Soyuz launch were carried out from Europe's Spaceport in 2021, placing a total of 21 payloads in orbit ranging from 1 kg to 6.4 tonnes (38.6 tonnes of payload in total), and representing commercial as well as institutional customers. European institutional payloads injected included two new Galileo satellites bringing the total number launched to 28, three CERES payloads, Syracuse-4A for France's defence procurement agency, and NorSat-3 for the Norwegian Space Agency, as well as ESA CubeSats SunStorm, RadCube and LEDSAT.

Of particular importance was Vega return to flight on 29 April (VV18) delivering the Pléiades Neo-3 Earth observation satellite and five auxiliary payloads into separate orbits using the Small Spacecraft Mission Service (SSMS) multiple payload adapter.

Lifting off on Christmas morning, an ESA-provided Ariane 5 rocket successfully launched the NASA/ESA/CSA James Webb Space Telescope. Owing to the high launch precision and resulting accuracy of the launch trajectory, Webb was able to save enough extra propellant to support science operations for a lifetime of up to 20 years, more than the five-year minimum baseline for the mission.

In September, the ESA Council agreed on the boundary conditions for the first three years of stabilised exploitation of Ariane 6 and Vega-C. The revised stabilised exploitation model is based on European institutional launch service demand for four Ariane 6 vehicles and two Vega-C per year on average. Price conditions for the associated standard launch services were also agreed upon.

Development (Ariane 6, Vega-C and Space Rider)

Preparations and testing for the Ariane 6 maiden flight continued throughout 2021, including the first combined tests at Europe's Spaceport in French Guiana, confirming the operations and equipment required for integration of the upper part of the launch vehicle under launch campaign conditions, shipment of the test specimens for the combined tests and the finalisation of preparations for the Ariane 6 Hot Firing tests of the Upper Liquid Propulsive Module (ULPM) at DLR. A notice of interest for an opportunity to fly payloads and experiments on board the first flight of Ariane 6 was published in November.

In preparation for the Vega-C maiden flight in 2022, the first elements of the first flight model of the new-generation launch vehicle reached Europe's Spaceport. Launch complex activities for the maiden flight are also progressing. A contract to advance Vega-C competitiveness to widen its mission capabilities was signed on 15 December.

Space Rider flight segment design activities progressed and will be followed by the system-level critical design review (CDR) set for April 2022. Three landing sites have been identified for the maiden flight from Europe's Spaceport, in Santa Maria Azores and Italy. An announcement of opportunity to fly payloads on ESA's Space Rider was published in October.

A development contract was signed on 13 July with prime contractor ArianeGroup to develop the Astris kick stage to enhance the versatility of Ariane 6, enabling it to offer a range of new space transportation services by allowing complex orbital transfers. ESA's Hera spacecraft, a planetary defence mission to the Didymos asteroid system, is set to be the first to use Astris.

A contract for Vega-E preparation was signed with Avio on 21 July. Vega-E will replace the third and fourth stage of Vega-C with a newly developed third stage to increase payload mass and volume and reduce launch service cost.

Future Preparation

Under the New European Space Transportation Solutions (NESTS) initiative, ESA issued several study contracts to define the elements of future European space transportation solutions. The studies will inform the preparation of proposals to be submitted for decision at the 2022 Council meeting at ministerial level.

In support of commercial space transportation projects, a follow-on contract under ESA's Boost! programme was signed with Isar Aerospace on 23 November for flight demonstrations of its Spectrum

launch vehicle planned to take place in 2023. Other contracts were signed with Orbex, Skyrora and D-Orbit for end-to-end space transportation services.

ESA forged ahead with advanced developments in two flagship space transportation demonstration projects, signed contracts for the Prometheus ultra-low-cost reusable rocket engine demonstrator and for the highly optimised Phoebus upper stage. This will benefit Europe's new Ariane 6 launcher in the near term and prepare for a new generation of European launch vehicles in the next decade.

Further advancing towards the demonstration of a reusable first stage vehicle, several test tank cryogenic filling and draining tests were conducted under the Themis programme. Tests of light-weight rocket tanks from carbon fibre reinforced plastic, a technology that could be used on a future improved upper stage, were successfully concluded.

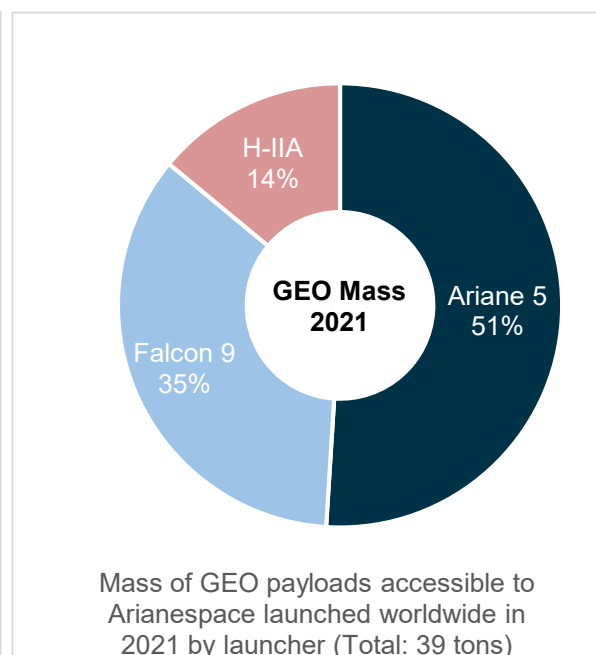
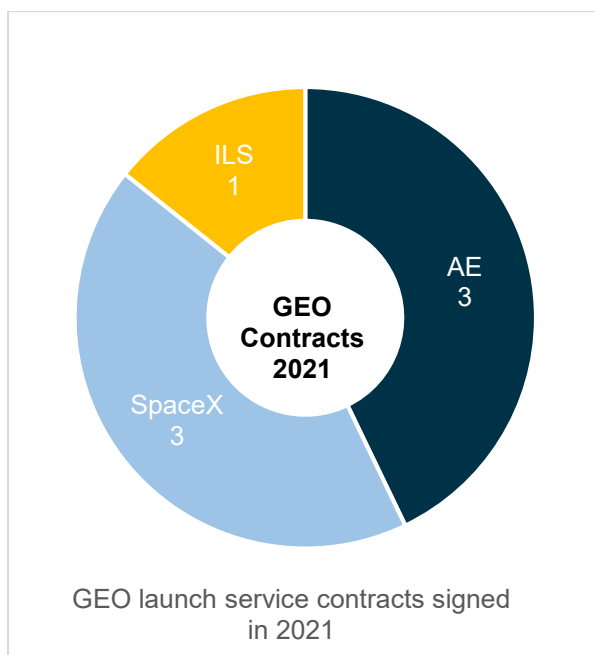
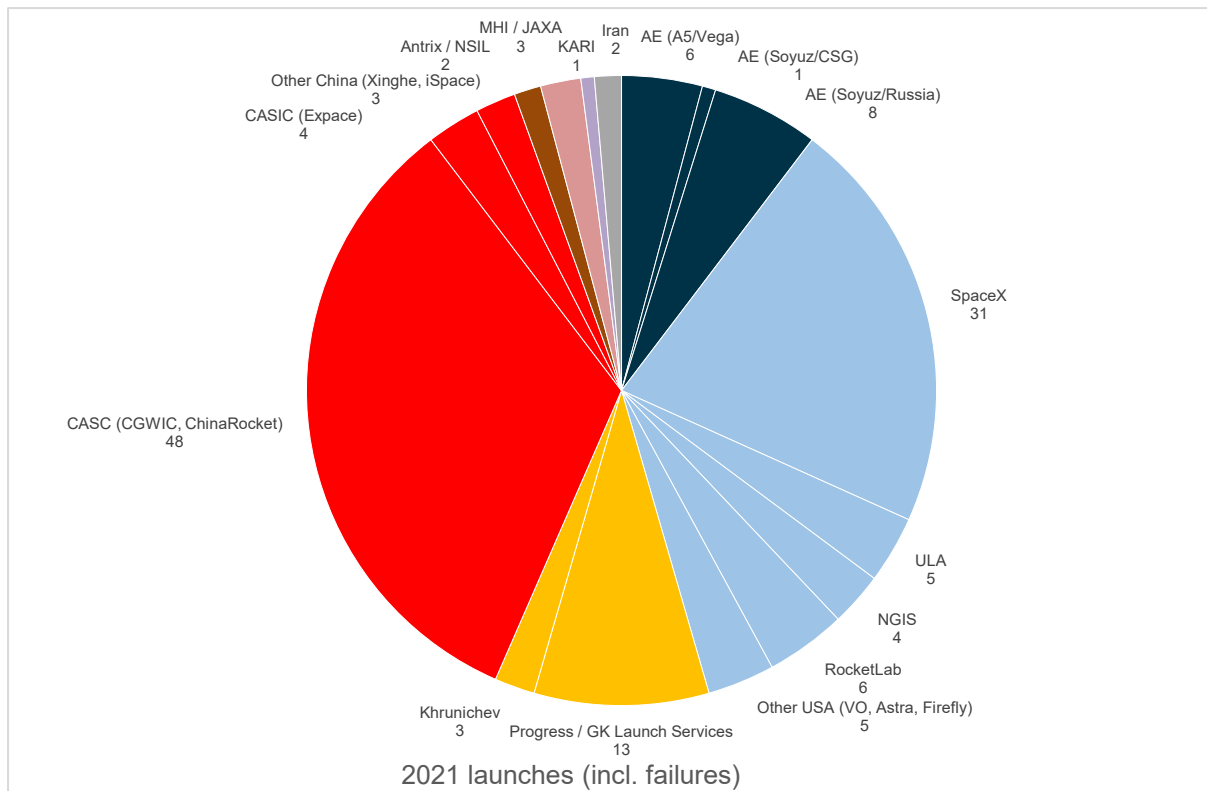
Europe's Spaceport

ESA invited European space and non-space companies to get involved in activities to modernise facilities at Europe's Spaceport in Kourou, French Guiana. A total budget of €140m has been allocated to finance this work over the 2020–24 period, including the construction of a new mission operations centre, installation of digital networks and preparations for an ambitious transition to sustainable energy sources. The launch range modification will also prepare the Spaceport to welcome additional launch service providers. A new additional radar called Amazonie-I was installed at Pariacabo, a high point at Europe's Spaceport and was tested during VA254 Ariane on 30 July.

The ELA-4 launch complex built for the upcoming Ariane 6 rocket, designed to optimise the launch system performance and operations workflow, was inaugurated at Europe's Spaceport on 27 September.

The Russian-built Soyuz rocket celebrated its tenth anniversary for launches from Europe's Spaceport, with the initial launch taking place on 21 October 2011 deploying the first pair of European Galileo satellites. By the end of 2021 there had been 26 Soyuz launches from Europe's Spaceport.

Facts and Figures



Operations

Despite the unexpected challenges posed by the pandemic, the Directorate of Operations (OPS) successfully executed various high-profile events in 2021, such as:

- The unique double encounter with Venus made by Solar Orbiter and BepiColombo just 33 hours apart which made space history. This offered an unprecedented opportunity to study the Venus environment from different locations at the same time and, moreover, in locations that are not typically visited by a dedicated planetary orbiter.
- The dramatic rescue of the Integral observatory following a reaction wheel failure, where the Integral Flight Control Team — together with Flight Dynamics and Ground Station Teams at ESA's ESOC mission control, teams at ESAC and Airbus Defence & Space — rescued the mission from its Emergency Safe Attitude Mode by implementing an ingenious new way to control the 18-year-old spacecraft.
- Successful support to NASA's Perseverance rover after its touch down on Mars. In order to communicate with Earth from its landing site on the Jezero Crater, the rover relied on ESA's TGO (Trace Gas Orbiter) and Mars Express orbiting Mars to relay data between Earth and the rover via a network of deep space stations, including ESA's Estrack network.
- The successful first encounter of BepiColombo with its target Mercury, and many challenges which have been overcome along the way. The mission will perform nine flybys in total to help steer it into Mercury orbit.
- The successful Solar Orbiter Earth flyby crossing the Geostationary and Low Earth Orbits littered with space debris. The Space Debris Office performed assessments based on Solar Orbiter's trajectory and the expected position of catalogued objects in orbit around Earth, providing a collision probability for any specific close approaches.
- The successful support to the James Webb Space Telescope (JWST) launch on 25 December. ESA's Estrack cooperative network, the 10-metre antenna in Malindi, Kenya, made first contact from the ground with the fledgling mission, with the all-important "first acquisition of signal". The first commands to JWST were sent by Malindi, which continued delivering telemetry and up-linking commands, ranging and Doppler until end of track.
- In December, ESA's Mars Express operations team carried out a series of data relay experiments with the Chinese Zhurong Mars Rover. The ESA orbiter and Chinese rover are unable to establish two-way communications due to incompatible radios. Nonetheless, the teams successfully demonstrated that Mars Express could receive signals transmitted "blind" from the rover as it passed overhead.
- The determination of a new "space time" which is measured by two atomic clocks at the ESOC mission control centre. The new "ESOC time" brings wide-reaching operational benefits for all ESA missions, making new feats possible in space while adding to our global definition of "now".
- The suite of ESA-operated missions including interplanetary, astronomy, Sentinels and the Earth Explorer satellites, together with the PROBA missions have been operated nominally, thanks to the excellent operations teams' adaptations in response to the pandemic.

Beyond nominal operations, preparations for the many upcoming missions included satellite validation tests and ground segment preparation. The operations team performed more than 20 critical avoidance manoeuvres, reacting to numerous collision avoidance warnings.

ESA Director General Josef Aschbacher and Director of Operations Rolf Densing signed the agreement between ESA and the Indian Space Research Organisation (ISRO) for the exchange of navigation, operations and data handling support. Estrack's global nature makes ESA an important partner for other space agencies that need to ensure reliable communications with their deep space missions, particularly during mission-critical phases such as LEOP. The first missions to benefit from the new cross-support agreement will be ISRO's Aditya-L1 solar research mission and Chandrayaan-3 lunar mission, both due for launch in Q2/Q3 2022.

Basic Activities

Along their three objectives the Basic Activities are pleased to highlight their most prominent achievements in 2021:

Enable the future via early elements of the seamless grid of innovation

In the **Discovery Element**, five new OSIP thematic campaigns were launched collecting novel early R&D ideas from industry and academia (over 1,500 ideas for Discovery and Preparation). The first ever ESA-wide call for ideas for new mission concepts received over 200 ideas, grouped and discussed in ten thematic workshops and resulting in 12 new pre-Phase A mission activities. A short description of all newly selected Discovery activities can be found on the main ESA website (https://www.esa.int/Enabling_Support/Preparing_for_the_Future/Discovery_and_Preparation/Idea_s_selected_through_the_Open_Space_Innovation_Platform).

The **Preparation Element** included in 2021 pre-Phase A and Phase A system studies that have been supported through the CDF (Concurrent Design Facility) for projects such as Sentinel-2 and Sentinel-3 Next Generations, PhiSat-2, the European Large Logistic Lander EL-3, Small Mars missions, Harmony/Stereoid, the Lagrange mission, and feasibility studies such as on space-based relativistic PNT (Positioning, Navigation and Timing) systems.

The **Technology Development Element** closed 52 activities and initiated 123 in 2021. The Work Plan budget for 2021/2022 committed a significant number of industrial contracts. It included 209 activities, 94 of which are in generic technologies and had a special emphasis on COTS & Digital Engineering.

Develop and maintain ESA's common infrastructure

Regarding **Mission Operation Infrastructure** activities, the second deep space antenna development in Australia in New Norcia (NNO-3) was kicked off. This will ensure maximum science data return from ESA's deep space missions. In ground segment development the European Ground Operating System (EGOS) Multi-Mission project was kicked off. Ongoing ground systems developments are on track. These include the next generation of operational simulator infrastructure, the next generation ground station monitoring and control system, and the new Orbit Facility for mission analysis and flight dynamics operations. These activities not only replace obsolescent software but also add new functionalities needed by future missions while also contributing to ESOC's portfolio of freely licensable mission-proven and operationally trusted software

ESA laboratories and the Test Centres have evolved to meet the requirements of projects and missions to ensure their effective and efficient delivery in progress. Major failure investigations,

qualification and testing campaigns were completed both at equipment and system level for major ESA projects, including Galileo, JUICE, Plato, Euclid, MTG, Sentinels and Vega.

In the domain of **Cyber security** the Security Cyber Centre of Excellence achieved important project milestones in 2021.

In the domain of **Common IT**, several applications were implemented such as the ESA Agenda 2025 Dashboard, esacontact (CRM), Digital Subscription Tool, Lotus Notes migration to Office 365. Also further improvements for the Corporate and Cyber Security IT Infrastructure were achieved.

With more than 60 active projects in 2021, **Estates and Facilities Management** focused on mitigating the Agency risk stemming from the backlog of obsolete infrastructures, installed energy-efficient equipment in support of the Director General's ESA Agenda 2025 and upgraded to new working environments. HQ refurbishment is on schedule for completion by the end of 2022.

Develop, preserve and disseminate knowledge, competences, and skills for capacity building and sustainable growth, inspiring and promoting creativity

The **ESA-wide Heritage Space element** continued to complement and support other ESA programmes and international initiatives by ensuring preservation, access, curation and exploitation of data from 140+ ESA heritage space missions/campaigns (one of the largest space data archives in the world), and of the ESA Records Archives.

In the domain of **Knowledge Management**, the repositories in the activities "Knowledge from projects, people and Information and Data" were expanded.

For the **Education** activity, the Education Office pursued the development of Space for Education 2030, ESA's long-term education programme to meaningfully support the implementation of the five strategic priorities set out in ESA Agenda 2025: by attracting and retaining the best talents, to remain competitive; and by supporting society in providing a long-term response to the emerging global challenges through education.

ESA's **Core Technical Expertise** has been focussing on supporting and strengthening internal competences in emerging technology fields such as digital engineering, artificial intelligence, and cyber security. An Automation and Artificial Intelligence Roadmap for Operations has been developed as well as a prototype to develop AI/machine learning competencies based on an Agile working method.

In the field of **Standardisation**, numerous standards and handbooks have been published as European Norms and Technical Reports respectively by CEN. The ECSS 4.0 simplification activity has completed the review of management standards.

For **Harmonisation**, further improvements for the ESA-EU 2020 European Space Technology Master Plan have been taken on board for the 2021 edition. Further OPS-relevant Technology Harmonisation Dossiers were published.

The **Earthnet** element continued to act as anchor customer for European New Space missions: the ICEYE satellite constellation was on-boarded as a new ESA Third Party Mission in 2021. Earthnet ensures data access to 50+ non-ESA missions for research, including operations support, with a steadily increasing scientific user base. International Charter activations reached a new height in 2021 with close to 50 activations.

Measures have been implemented in the **Technology Transfer** and **Incubation** area to increase Member States' industrial competitiveness. The Business Incubation Network could now be expanded

to 22 Centres with representation at more than 70 locations across nearly all ESA Member States and support of more than 1,100 companies. Additionally, a new activity, Space Solutions Alliance, was established.

The **SME Initiative** has progressed further towards easing the way for SMEs to do business with ESA including the SME web portal, the ESA Learning Hub (see <https://learninghub.esa.int>) and the development of “esa-match”, a matchmaking platform for industrial entities.

Technical and Quality Management

The Directorate provides hands-on engineering skills and specialist expertise to all project teams across the Agency. In 2021, after the first introduction in Euclid and then in PLATO project, Model Based System Engineering (MBSE) has continued to be introduced more systematically in CDF (Concurrent Design Facility) studies and industry (e-Deorbit, Adrios, e-Inspector, Ariel, Envision, EL3, MSR-ERO, G2G).

General Support Technology Programme (GSTP)

Examples of achievements for 2021 include:

The Low-temperature Near-field Terahertz chamber integrated in ESTEC to meet JUICE needs completed eight days of cryogenic radio-frequency testing for SWI (Sub-millimetre Wave Instrument) for ESA’s JUICE mission.

The new National Space Propulsion Facility in the UK was declared open. GSTP has invested around €4.5m and overseen the design, assembly and commissioning of the facility, equipped to test-fire the most powerful classes of rocket engines used on board spacecraft.

Two ESA CubeSats developed through GSTP’s Element 3 – Fly were launched. SunStorm, an activity with Finland, and RadCube, an activity with Hungary, will demonstrate miniaturised space weather instruments for use in later operational space weather missions. SunStorm, a two-unit CubeSat, hosts a new solar X-ray flux monitor that will detect coronal mass ejections from the Sun which threaten satellites and terrestrial power and communications networks.

Small missions developed in TEC received international attention in 2021 as well.

PROBA-1, still in orbit after 20 years while it was intended to survive just two years, provided 1,000 images of more than 1,000 sites. This has been vital for monitoring several environmental concerns, from assessing how different land use strategies in Namibian savannah have affected vegetation growth, to vegetation types in Central Namibia’s savannahs, or helping to map and understand alpine snow cover in Swiss national parks.

The Hera mission closed the system and all subsystem preliminary design reviews in 2021. Collaboration with NASA’s DART investigation team has been maintained with several Hera team members participating in DART’s science data preparation reviews. The DART spacecraft was successfully launched on November 23 from Vandenberg, California and is currently en route to asteroid Didymos. Impact is planned on 26 September 2022.

Several activities were undertaken in support of fighting the COVID-19 pandemic.

Medical supplies and samples were delivered by drones to four National Health Service (NHS) centres in Scotland. They were enabled by space-based technology SEDDCR (GNSS waypoints, SATCOM for

coverage across entire route, EO data for emergency trajectory modelling) as the result of a pilot test phase. The initiative improved the care of 733 patients during trials, saved 10,300 hours in time and 11,000km were flown in 355 flights.

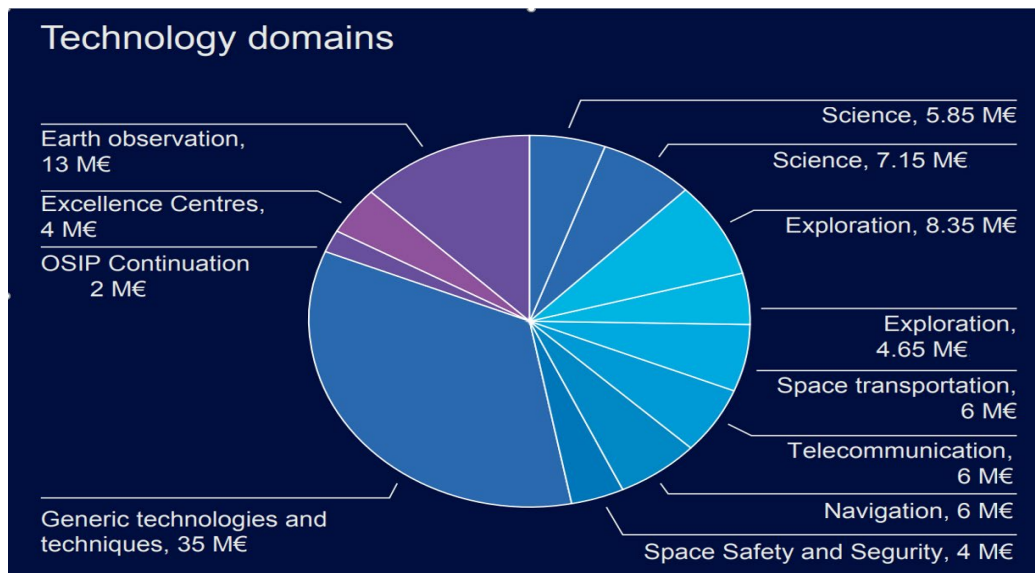
Heritage from B-Life for EBOLA crisis (2015) was used to support the Piedmont region (IT) in June 2020 (no rapid testing nor vaccines being available back then) for COVID-19 rapid screening and epidemiological mapping of 25,000 members of the healthcare workforce and blue forces. The WHO and European Commission expressed interest in knowledge and experience transfer in emergency management from that project.

Facts and Figures

Technology Development Element (TDE)

By testing the feasibility of prospective technologies, the Technology Development Element (TDE), which is a mandatory programme supporting all ESA's fields of activity, gives Europe the ability to look ahead to plan and define future space missions and activities, helping to "future-proof" the European economy.

In the framework of the programme, ESA's Technology Strategy is implemented on a regularly updated 2-year rolling work plan. This basic and mandatory programme has a work plan budget of €102m and average annual commitment (industrial contracts) amounting to €55m. At least 150 new activities are initiated per year. Projects are undertaken by European academia and industry under ESA supervision. Recent examples have included a 30% efficient spacecraft solar cell, an air-breathing engine for low-flying satellites, parachute designs for Mars and a laser tuner for advanced instruments.



TDE Spending per Technology domains

TECHNOLOGY AND ENGINEERING TARGETS



30% IMPROVEMENT
OF S/C DEVELOPMENT
TIME BY 2023

We develop key
technologies
to allow ESA to reduce
the time from Phase B2
to launch

ORDER OF MAGNITUDE
BETTER COST
EFFICIENCY

We develop key
technologies
to allow Europe to achieve
cost efficiency
improvements
with every space system
generation

30% FASTER
DEVELOPMENT
& **INNOVATIVE**
TECHNOLOGY ADOPTION

We develop processes,
methods and technologies
to allow Europe to fully
benefit from the early
introduction of new
technologies

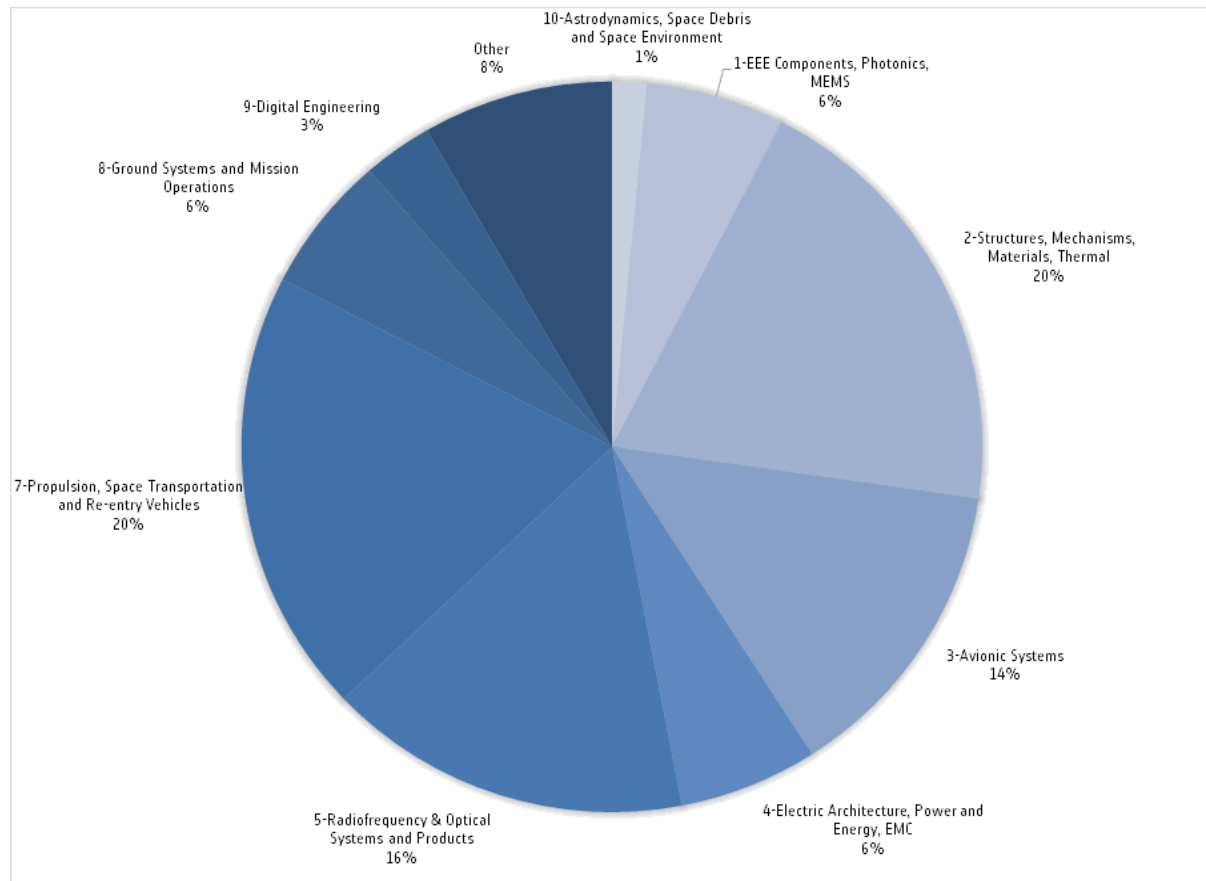
INVERT EUROPEAN
CONTRIBUTIONS
TO SPACE **DEBRIS**
BY 2030

We develop technologies
to leave a safer space
environment than we
inherited to the next
generation

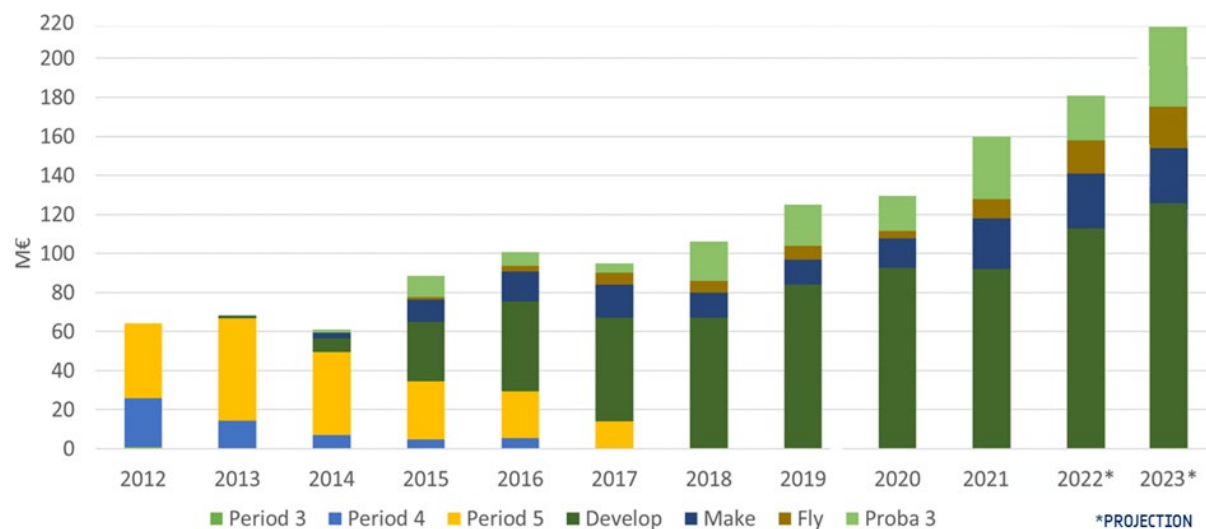
2020 PRIORITIES: DIGITALISATION & COTS

General Support Technology Programme (GSTP)

Through the optional General Support Technology Programme (GSTP) ESA, Participating States and industry work together to convert promising engineering concepts into a broad spectrum of useable developments. The GSTP Annual Commitment for 2021 amounts to €131m, with 132 activities being completed and 157 initiated.



Percentage of GSTP activities per competence domain for 2021



Projection of growth for GSTP till 2023

ESA Open Day at ESTEC 2021

Two successful separate sessions of the ESA Open Day took place on 2 and 3 October on the theme of “Inventing the Future”. On 2 October, a hundred disabled guests visited ESTEC where sixty volunteers from all Directorates had organised demonstrations for them. The event was a great success drawing an enthusiastic response from the visitors. On 3 October, more than 1,100 registered viewers attended a virtual ESA Open Day animated by 80 ESA speakers and moderators in the presence of ESA astronaut Alexander Gerst. Viewers were from 78 countries inside and outside Europe with more than 5 million people following on Twitter. Online ESA Open Day with live presentations from different establishments featured One ESA and gave the opportunity to viewers from non-Member States to attend.

ESA Open Day for people with a disability – 2 October 2021



Facts and figures

- **160 registrations**
- 99 visitors on site
- 61 ESA volunteers from all Directorates
- TEC, EO, SCI, HRE, OPS, STS and HR were presented

Conclusion

- It was a big success for a pilot event
- Visitors were very enthusiastic and grateful
- Other institutions have been in contact to require if the similar ESA Open Days will be organized in future
- Long list of lessons learned

Quote of a visitor

- “I was treated as a normal person”

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2



European Space Agency

Transversal approaches

Downstream Gateway

The *Downstream Gateway* has changed its name to **Commercialisation Gateway** to better reflect the enlarged scope of its activities in the CIP (Commercialisation, Industry and Procurement) Directorate in line with ESA Agenda 2025 pillars. In preparation for CM22 and in support of ESA programmes, several market studies have been conducted to understand trends, needs, new business opportunities and finally to support the decision-making process. The verticals and use cases investigated are: Investments, Green & Digital Finance, Insurance, Additive Manufacturing, IoT, Smart City, Renewable Energy, Offshore Energy, Videogaming, Education, Cultural Heritage, Tourism, Sport & Equipment, Health, Connected Hospitals and Biopharma. The methodology adopted leverages on intense desk research supported by structured interviews (more than 200 in number) with key stakeholders and ESA experts.

Based on the market analysis, macro-economic considerations and programme needs, a series of actions have been initiated to connect and engage with key stakeholders and to stimulate demand and new market opportunities:

- Reach-out actions in areas with great potential of disruption and/or acceleration of current innovation, resulting in the organisation of a series of webinars and workshops on Digitalisation, Air Quality, 5G, Urban Planning, Green and Sustainable Cities, Digital Health, Cultural Heritage, Green Finance and In-orbit economy. This was done through the engagement of important old and new actors in each sector from industry, institutions, academia, and the investment sector as well as ESA experts and management. Follow-up actions have been initiated wherever these are of interest for ESA programmes.
- Definition and implementation of an aggressive digital marketing strategy leveraging social media, newsletters, and website promotion (<https://commercialisation.esa.int>).

Multi-Directorate Activities

The concept of multi-Directorate activities, going beyond the traditional boundaries per domain, was elaborated and discussed with the Member States. A certain number of such activities were identified, where putting together the forces coming from different Directorates can lead to a new type of proposal, serving better the needs of the potential users. This is further elaborated in 2022 and will lead to specific proposals proposed for subscription at the next Council at Ministerial level.

Digital Transformation

Digital Agenda

ESA's digital transformation is targeted in such a way as to boost the Agency's efficiency and effectiveness, streamlining processes, modernising the workplace to attract the best talents, and establishing a data-driven organisational culture, treating data as a central and critical corporate asset to protect, measure, analyse and improve.

Building on its past digitalisation achievements, the Agency was able to achieve uninterrupted operations throughout the pandemic. The workplace has been streamlined and enhanced with the deployment of modern collaboration tools such as Office365 and Teams. Multiple IT service and infrastructure improvements have been introduced to support secure remote working for all.

Digitalisation of Agency core processes has continued in Human Resources (finalising performance reform), Finance (automating numerous processes), Procurement (functionalities to improve efficiency) and Industry Support (esa-match).

As part of the ESA data strategy and data-driven culture following on from the achievements of the data lake project developed in 2021, Agency-wide dashboards have been developed for Council (Agenda 2025) or are being developed (ESA dashboard) combining with analytics online financial, procurement, geographical return, workforce and other key metrics to support faster more informed decision making.

Consolidated software repositories and increased use of common and virtualised computational resources are allowing a better use of resources and higher availability of data. With an ever-increasing variety and volumes of space data, high performance computing (HPC) technologies are being studied for their potential to solve computational challenges in performing complex simulations across ESA programmes. New capabilities planned for inclusion in the Basic Activities proposal to Member States to transform big data with artificial intelligence (AI)-based data analytics into smart information products will provide societal benefits in view of the European Green Deal and Digital Agenda.

Information Technology

The ESA IT Department has delivered a new baseline that enables and maintains new ways of working that respond to changing requirements that have occurred during the COVID-19 pandemic. The IT Department continues to support and improve the teleworking experience. They have released the new computing environment and tools for ESA colleagues that improve and increase opportunities for digital collaboration both internally and with external stakeholders.

New features: Sharing Agenda 2025 Progress

The IT Department partnered with internal stakeholders to create and launch the Agenda 2025 dashboard at the Intermediate Ministerial Meeting in November 2021.

Upgrading our ways of working: esa365

The ESA IT Department deployed 30% of the esa365 (IT solution family with the integration of Microsoft Office 365) migration to the entire workforce in 2021.

Supporting ESA's Core Business through digital administration tools

ESA's human resources tool hr-space has been adapted to include new rules for flexibility and teleworking. This update and several changes in the performance module was implemented within the constricted timeframe. Visual-esa — a whiteboard tool — is ESA's new visual lean management solution. Among those participating in the project was astronaut Alexander Gerst.

Cybersecurity for ESA

Cyber security measures have been reinforced by the ESA IT Department, as ESA continues to proactively improve its cyber security posture.

ESA's IT Infrastructure

A new network infrastructure and security service contract has been put into place. This will allow a full transition of ESA's network and related security services to an infrastructure in line with the current state of the art.

Resources Management, Corporate Activities and External Relations

Human Resources, Facility Management

Human Resources: Diversity and Inclusiveness

In 2021, in the framework of the Diversity and Inclusiveness initiatives, ESA reinforced its commitment in attracting and retaining a gender-balanced and diverse pool of talents. Diversity and Inclusiveness are key priorities to secure the fulfilment of the ESA Transformation envisaged in the Agenda 2025. The Agency has especially reinforced its commitment to gender balance, having achieved the EDGE certification for gender equality, in order to measure its position against objective criteria and benchmark its initiatives against other organisations. ESA has been striving to reduce visible and invisible barriers and ensure an inclusive working environment to people with disabilities; the Parastronaut feasibility project has given the Agency further visibility as a role model with respect to disability inclusion.

Estates and Facilities Management

The Estates and Facilities Management (EFM) Department continued to confront the twin challenges of managing many of the practical consequences of the COVID-19 pandemic while facilitating the core business of ESA. A rapid, flexible response was essential as the waves of infection impacted ESA's widely distributed sites with variations in time and intensity. The overriding objective was to enable onsite staff and contractors to perform effectively in the safest possible working environment. Working closely with the Director General centrally and heads of establishment and centres locally, a careful balance was struck at all times. The positive trends in the pandemic in mid-2021 were rapidly followed by restrictions imposed by the omicron variant. Through all this, ESA's sites remained open, allowing critical tasks to be performed and high priority activities supported.

The Department's core role of maintaining and investing in ESA's key supporting infrastructure continued within a particularly challenging commercial context. Upward price pressures driven by European COVID-19 recovery measures impacted investment activities across our sites and necessitated the reissuing of a number of tenders and the most careful consideration of scope. Another noticeable impact was the difficulty of ESA's FM contractual partners to recruit and retain the skilled workers that are an essential component of their services. Inflationary pressure was noticeable in all areas of the Department's activity, compounded towards the end of the year by a surge in utilities prices.

Throughout the year a wide range of projects were implemented and FM contracts amended to ensure that ESA Agenda 2025's high ambitions for an environmentally sustainable agency met with a full response. The highly complex ESA Headquarters renovation project reflects these ambitions and 2021 saw significant progress in that regard. While COVID-19 will have some impact on the project timeline, this has been limited by management action and the other key project performance indicators remain unchanged; ESA will return to its rebuilt HQ in early 2023.

Commercialisation, Industry and Procurement

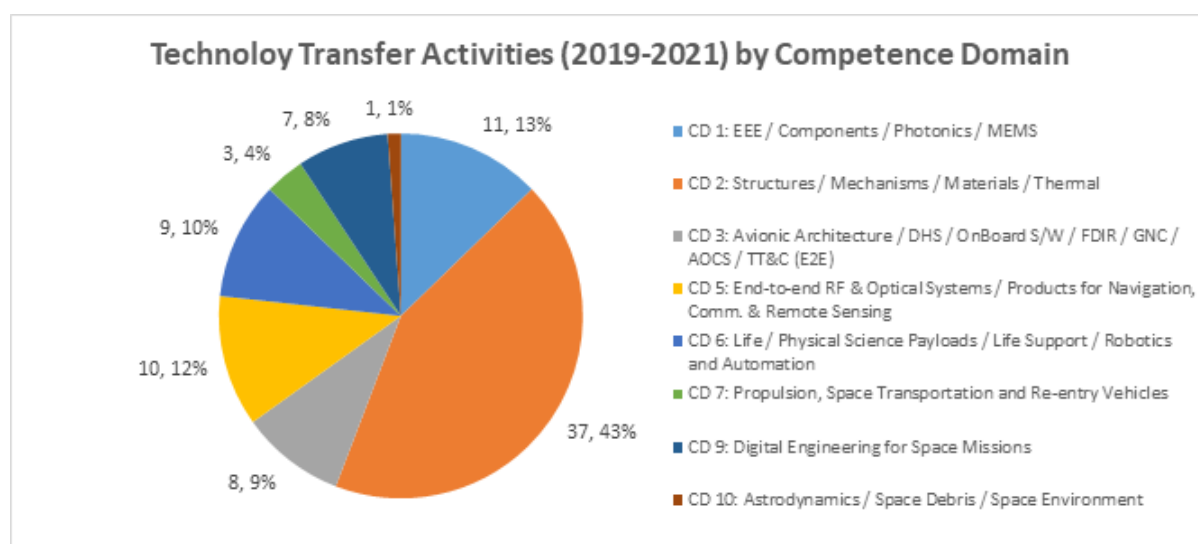
Commercialisation

Transfer of Technology and Patents

Ensuring that a technology is transferred, i.e. used for a new space or non-space application, will ensure that knowledge is disseminated and ultimately contribute to the generation of new commercial opportunities and public benefits. Funding, expertise and network access are offered to industry for technology and commercial maturation thereby facilitating the technology transfer process in a comprehensive manner.

The Technology Transfer and Patent Office is running eight new Technology Transfer Demonstrators and eleven Technology Transfer Feasibility and Design projects. The selection of six additional Technology Transfer Feasibility and Design projects is ongoing.

Forty patents were granted in 2021 in different countries around the world, fifteen new invention disclosures were received and processed by the Secretariat, while five new patent applications were filed. Two Commercial License Agreements for space use and two Research License Agreements for non-space were signed.



SME office

The SME Office continued its activities for the implementation of the SME Policy. The Agency's new matchmaking platform, esa-match, was launched in March to support registered entities by facilitating partnering, teaming and cooperation. In December, the online edition of the ESA Industry Space Days with a focus on commercialisation was organised with 1,400 participants from 40 countries and 850 organisations attending. In 2021, training courses for SMEs were organised with 526 participants from 349 companies.

ESA BICS and Brokers

The Section responsible for the ESA BIC and Broker network has moved to the newly established Directorate of Commercialisation, Industry and Procurement (CIP). By providing access to the talent of thousands of European entrepreneurs, ESA BICs have become a fast and efficient way of harnessing the potential of space technology and space-based applications across all sectors of industry. The ESA

BIC network now consists of 24 ESA BICs managed under 20 contracts, with representation at more than 70 locations across nearly all ESA Member States. More than 1,150 companies have so far been supported. The ESA Technology Broker Network is responsible for enhancing and accelerating the innovation ecosystem in the various Member States, by facilitating commercial opportunities that stem from technology transfer between industries as well as the utilisation of space infrastructure and its data. This support is transversal across both the upstream and downstream segments.

Industrial Policy and Audit

ESA continued in its efforts to:

- streamline geo-return management and calculation processes, paying particular attention to the forecasting and reporting/dashboard areas;
- ensure the development of industrial capabilities, especially in under-returned Member States;
- measure the space economy and assess the socio-economic impacts of ESA programmes.

257 audits were performed involving 120 companies to determine cost impacts due to COVID-19 in 20 major projects. Industrial audit activities reported €220m savings for ESA (€450m over 2018–2020). 2021 saw Lithuania become the third ESA Associate Member while Cyprus and Bulgaria agreed to continue their cooperation through renewed PECS+ agreements.

Procurement

One of the main objectives for the Procurement Department in 2021 was to contribute to ESA-EU FPPA (Financial Framework Partnership Agreement) and CA (Contribution Agreement) negotiations.

In addition, actions were initiated with a view to implementing the relevant parts of ESA Agenda 2025, namely:

- stimulating a more vibrant and dynamic commercial space ecosystem to help bring about the green and digital transformations;
- boosting commercialisation by facilitating rapid innovation and encouraging a different approach to risk-sharing in areas with commercial growth potential;
- updating the overall procurement approach and significantly reducing the average time-to-contract for contracts below €1m.

In March, the new esa-star Publication module went live, replacing the previous EMITS (Electronic Mail Invitation to Tender System) application which had been functioning since the end of the 1980s. The system ensures the publication and dispatching to economic operators doing business with ESA of tender actions, news, administrative and regulatory documents and more.

In 2021, major contracts were signed such as the Implementation Phase of the Ariel mission and the implementation phase of the Ariane 6 and P120C Transition Programme.

In addition to many “classical” contract management activities, 2021 has seen the implementation of innovative service development approaches such as, in Earth observation, the contracts for Arctic Weather satellites, two SCOUT projects and, in Human and Robotic Exploration, the signature of the Commercial Lunar Mission Support Services and the organisation of the Space Resources Challenge for the Development of Prospecting Technologies.

Finance, Planning and Control

The Department continued to operate in 2021 in a hybrid onsite/remote working configuration.

Financial Accounting and Operations

The Department continued to operate in 2021 in a hybrid onsite/remote working configuration.

ESA is completing the closure of the 2021 accounts, aiming for a tenth consecutive "clear" audit opinion. This will help achieve continued stakeholder trust and confidence in ESA's financial management.

Through highly digitalised processes, ESA accurately managed and accounted for 37,000 payment transactions, incorporating special business continuity measures in support of industry during COVID-19, with an average time-to-payment of 13 calendar days compared to ESA's usual 30-day contractual terms. Digitalisation advanced in invoice processing.

Treasury management efforts focused on ensuring sufficient liquidity to cover payments while limiting negative interest costs.

Corporate Planning, Budgeting and Controlling

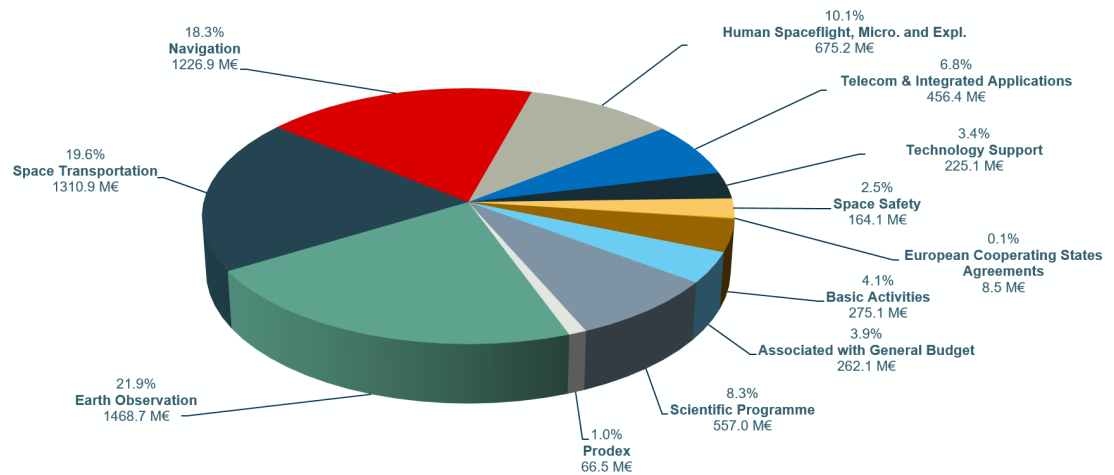
Planning and Controlling consolidated and assessed ESA's multi-year financial plans leading to endorsement of the 2022 cost plans by all Programme Boards and Committees and subsequent Council adoption of the 2022 budgets (for some €7 billion including third party activities).

Effective collaboration with Member States supported optimised income flows. The team supported the definition and implementation of the final FFPA terms agreed with the European Commission and EUSPA. Digitalisation initiatives progressed including the Digital Subscription Tool (DST) for Member States and the implementation of dashboards for the monitoring of execution against financial plans.

Income

ESA Mandatory Activities (including the General Budget and Scientific Programme) and activities associated with the General Budget are financed by Member and Participating States' contributions. ESA Optional Programmes are financed by Member and Participating States' contributions, and for some programmes, also by other entities, including the European Union and EUMETSAT. For Optional Programmes, Participating States declare a voluntary subscription on a multi-annual basis. Third Party Programmes, managed by ESA, are fully financed by third parties.

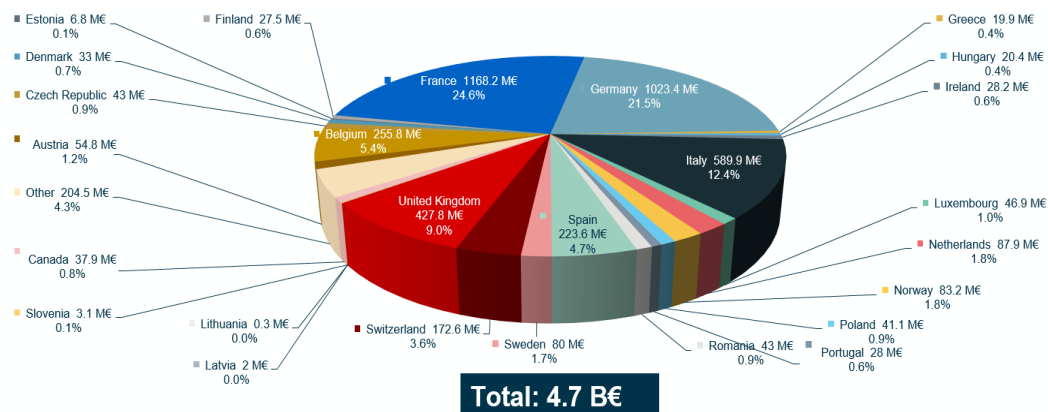
ESA 2021 Budget by Domain: 6.7 B€*



* Includes activities implemented for institutional partners

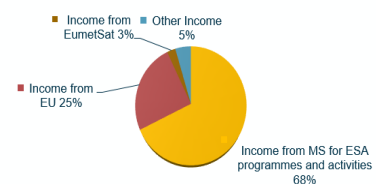
ESA 2021 Budget by Funding Source

Funding for ESA Activities and Programmes



Budget 2021 by Funding Source

Total: 6.7 B€



Cost

ESA's total cost in 2021 amounted to €5,776M, of which €3,943M related to ESA programmes and activities and €1,832M to Third Party activities.

Mandatory Activities represented 14% (€821M), 52% was accounted for by Optional Programmes (€2,985M), 2% by activities associated to the General Budget (€103M), 32% by programmes financed by Third Parties (€1,832M) and 0.1% by others (€35M).

The majority of cost accrued in 2021 by domain is provided below.

Basic Activities: Discovery, Preparation & Technology Development (41%), Mission Operations Infrastructure (13%), Engineering Labs & Test Centres (12%) and Sites & Common IT Investment (11%)

Scientific Programme: JUICE (31%), Missions in operation (17%), PLATO (13%) and Euclid (12%)

Space Transportation: Ariane 6 Development (25%), LEAP Ariane 5 Supplementary (14%), Common Infrastructure for Ariane 6 (8%) and LEAP Ariane 5 Classical & MCO (7%)

Earth Observation: Copernicus Space Component (33%), EOEP Period 5 (21%), MetOp Second Generation Development (13%) and MTG Development (9%)

Human and Robotic Exploration: Mars Sample Return (23%), Orion-ESM Flight Models 1 to 3 (17%), ExoMars (13%), Lunar Gateway (10%) and ISS Operations (9%).

Telecom and Integrated Applications: Core Competitiveness (43%), PARTNER (25%) and IAP/BASS (18%)

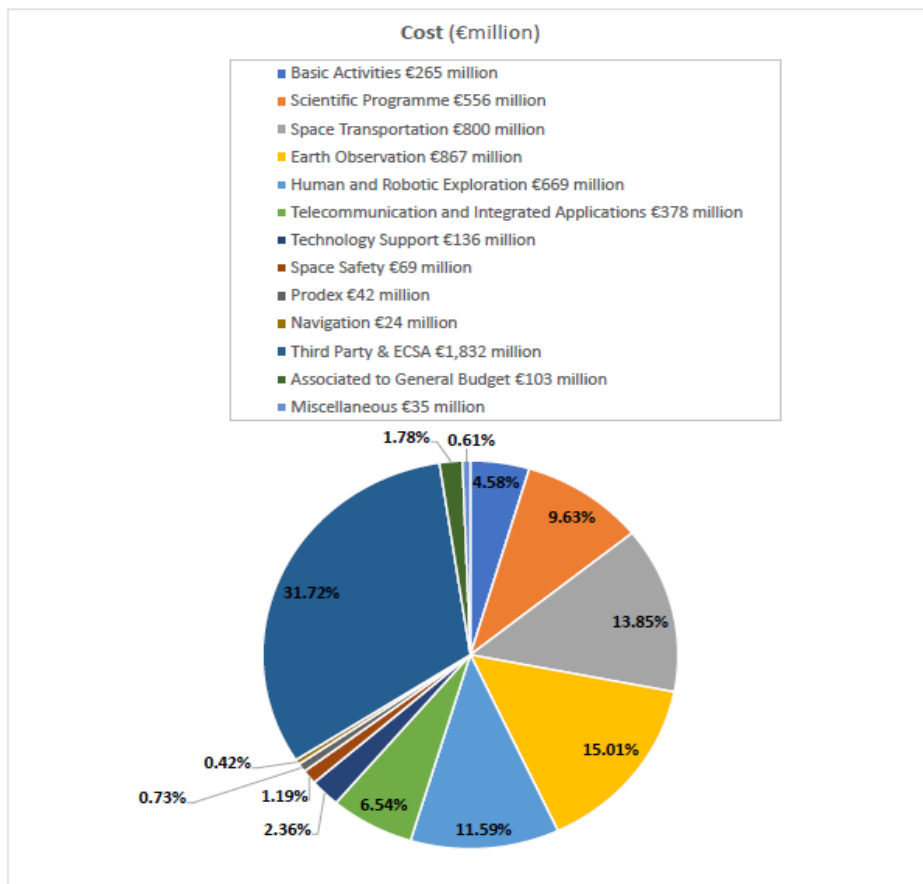
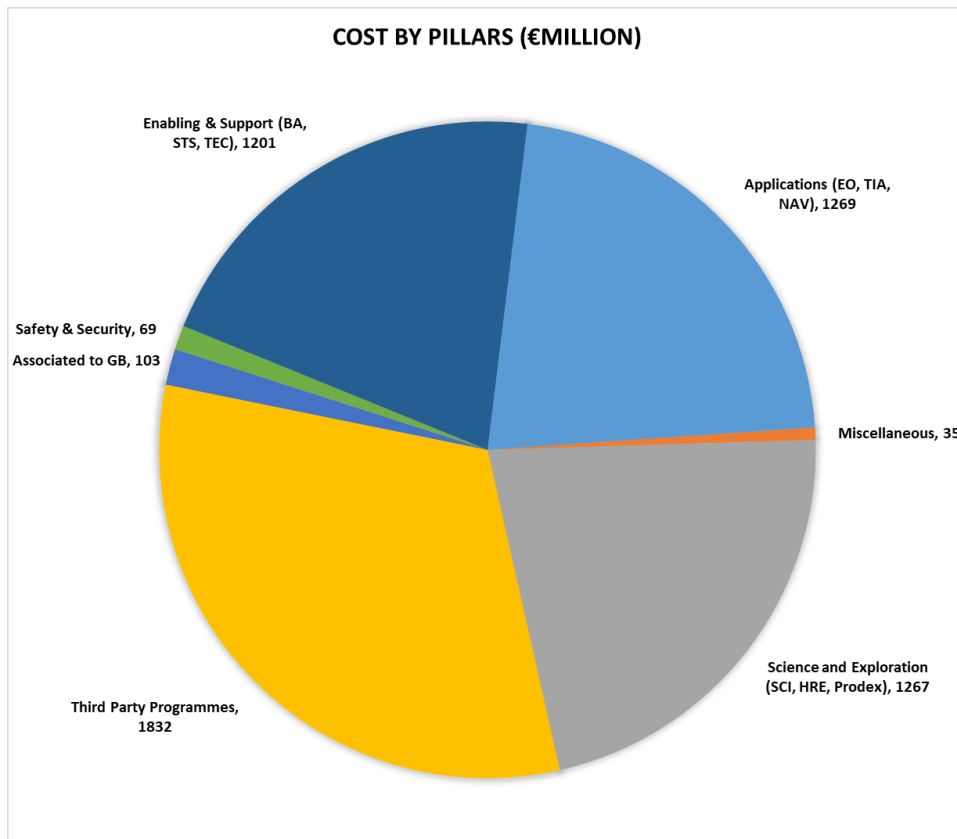
Technology Support: General Studies Technology Programme (86%) and Proba (14%)

Space Safety: HERA Mission (46%) and Core activities (21%)

Navigation: NAVISP (98%)

Third party: EU programmes, namely Galileo FOC DC - GSA – EUSPA (65%) in Navigation and Copernicus MFF (20%) in Earth Observation

Associated to GB: CSG 2020-2024 (87%)



Workforce Management

In 2021, a flexible utilisation of the staff ceiling further enabled the Agency to advance recruitments to address the ongoing departure wave and to manage workload peaks resulting from the activities decided on at Space19+. Despite the ongoing COVID-19 pandemic, recruitment activities gathered pace compared to 2020, reaching a level of 188 recruitments throughout the year. The contractor workforce, both loan employment and service, has remained stable. The Agency has intensified its actions to reinforce compliance with national labour legislation.

Member States	Hors					Total Staff
	Classe	A	L	B	C	
Austria	1	36	1	2		40
Belgium	1	105		4		110
Czech Republic		8				8
Denmark		11				11
Estonia		2		1		3
Finland		18		3		21
France	3	487	9	59	3	561
Germany	2	433	4	35		474
Greece		26		2		28
Hungary		9		1		10
Ireland		16		3		19
Italy	1	378		16		395
Luxembourg		5				5
Norway		24		2		26
Poland		31		1		32
Portugal		30		2		32
Romania		33		1		34
Spain		200		3		203
Sweden		39		2		41
Switzerland	1	24		1		26
The Netherlands	1	61		9		71
United Kingdom	1	245	3	19		268
Total	11	2221	17	166	3	2418
Associate Member States						
Slovenia		2		1		3
Latvia						
Lithuania						
Non-Member States						
Canada		23		1		24
Total		25		2		27
Grand Total	11	2246	17	168	3	2445

Legal Services

Two landmark results achieved in 2021 were the negotiation and successful conclusion in June of the Financial Framework Partnership Agreement and the accompanying Contribution Agreements between ESA, the European Union (EU) and the EU Agency for the Space Programme (EUSPA), together with the adoption of the Council Resolution on Accelerating the Use of Space in Europe (“The Matosinhos Manifesto”, ESA/C-M/CCCII/Res.1 (Final)) by the Ministers of ESA’s Member States in November. Beyond those milestones, the ESA Council took a large number of decisions, including the adoption of:

- a Resolution on the application of the common legal framework governing the exploitation of ESA developed launchers and Soyuz from the CSG (ESA/C/CCXCV/Res.1 (Final));
- a Resolution on the Agency's budgets for 2021 (ESA/C/CCXCV/Res. 2 (Final));
- a Resolution on the setting-up of a Council Working Group for the preparation of the intermediate Council meeting at ministerial level scheduled in the last quarter of 2021 (ESA/C/CCXCVIII/Res.1 (Final));
- a Resolution on the Financial Statements of ESA for the year 2020 (ESA/C/CCXCVIII/Res.1 (Final));
- a Resolution on the establishment of a Preference Clause as additional Industrial Policy Measure for the Period 2021-2024 (ESA/C/CCC/Res. 1 (Final));
- a Resolution on principles related to Ariane 6 and Vega-C exploitation models and new space transportation solutions (ESA/C/CCXCIX, Res.1 (Final));

Throughout the year, a total of ten international agreements were concluded with governments, international organisations and institutions of both Member States and non-Member States (see Annex 1), enabling international cooperation in space applications, human spaceflight and exploration, science, space transportation and other domains. In 2021, ESA also welcomed Lithuania as an Associate Member, following the ratification and entry into force of the respective Association Agreement. ESA’s Legal Services Department accompanied the drafting, negotiation and decision making in respect of those instruments, prepared a wealth of other legal documents for decision by Council and its subordinate bodies, and provided legal services for the functioning of the Agency and the implementation of its programmes and activities.

Communication

A new Director General, a new vision for ESA, and an omnipresent COVID-19 best characterise the context for communication in 2021.

A new #ESAAmbition campaign was implemented to communicate ESA Agenda 2025 and to position the new DG as its champion. A dedicated Chief News Officer team ensures that the Director General leads and directs ESA’s communication efforts, fully exploiting societal and political news trends. Internal leadership communication was enhanced by means of regular podcasts, vodcasts, and interactive formats to maintain a dialogue with the workforce while ESA performs its transformation.

Collaboration with ESA Member States in the framework of the Advisory Communication Committee is proving rewarding. Joining forces made the Astronaut Selection Campaign the most impactful communication opportunity in 2021. Efforts to translate all ESA media releases into French, German, Italian, Spanish and Dutch as well as using ESA WebTV to stream content from CNES, UKSA, ASI and DLR have significantly raised awareness.

To extend ESA's brand footprint among young and female audiences, co-branding projects have been implemented with Playmobil, Mattel and Coldplay - in addition to specifically targeted media relations. A brand licensing programme with Rocket Licensing Ltd began licensing the ESA brand to qualified manufacturing companies worldwide. Together with the online ESA Space Shop, the physical ESA Space Shops at ESRIN, ESOC and ESTEC provided for an enhanced brand experience in 2021.

ESA brand awareness is stable at 29% across all ESA Member States (55% Austria - 11% UK). 2021 was a good year with 70% more gross reach in external media. ESA Agenda 2025 was covered constantly throughout the year.

Relations with Member States

In 2021, the Director General regularly visited and exchanged with Member States, often taking the opportunity offered by various major events. This included meetings with Ministers and politicians in charge of space matters and related fields, with delegations, and addressing parliaments, industry, the media, or universities whenever possible. Most meetings were still held remotely, owing to the COVID-19 crisis. In addition, the Executive remained in close touch with relevant national counterparts.

External Relations

Relations with EU Institutions

In 2021, major milestones were achieved in extending relations between ESA and the European Union. The EU Space Regulation was published on 12 May with the largest allocated budget ever at €14.88 billion. Shortly after, on 22 June, the Financial Framework Partnership Agreement (FFPA) and the related Contribution Agreements were signed by ESA, the European Commission and EUSPA. The agreements signed represent an EU investment of almost €9 billion over the 2021–2027 period, for ESA and European industry to design new-generation systems and programmes, crucial for the economy and a green and digital Europe. This funding adds to ESA's budget and thus consolidates an ambitious set of mandatory and optional programmes, as defined by ESA Member States. The FFPA also includes additional components under ESA responsibility, such as the preparation and design of the future GOVSATCOM space segment and the validation of a prototype for a Quantum Key Distribution satellite in support of the new EU flagship on secure connectivity, In-orbit demonstration/validation service and space safety activities to monitor space hazards. Initial activities have been kicked off in the framework of these agreements for the various components. Among key milestones were the establishment of the European Commission Procurement Boards, the submission of the first Work Plans, the establishment of the Joint Office and the formal kick-off of activities.

International Relations

Cooperation with EU member states not members of ESA continued through Association (with Slovenia, Latvia and Lithuania, the latter joining in May, while Slovakia's application is being processed), reinforced European Cooperating State cooperation (Bulgaria, Cyprus), and general cooperation (Croatia, Malta).

In 2021, bilateral cooperation was successfully pursued with Canada through multiple cooperative activities.

The NASA-ESA-CSA James Webb Space Telescope lifted off successfully on an Ariane 5 from the CSG on 25 December. ESA astronaut Thomas Pesquet had a successful Alpha mission and Matthias Maurer followed him to the ISS in November with Cosmic Kiss. Multiple agreements were extended with

NASA. An agreement between ESA and NOAA on space weather and an implementing arrangement on cooperation on Lagrange/SWFO were approved.

ESA and Roscosmos continued their close cooperation in the joint ExoMars-2022 programme. Other 2021 highlights were the launch of the European Robotic Arm attached to the Russian Multipurpose Laboratory Module in July, delivery of the PILOT-D camera demonstrator for the Luna-25 mission and the successful launch of a Soyuz-STB with two Galileo FOC satellites from the CSG.

ESA and the Chinese Academy of Sciences implemented joint SMILE and Einstein Probe missions. ESA and the National Remote Sensing Centre of China launched the fifth edition of the joint Dragon programme.

The ESA-JAXA BepiColombo mission made several flybys on its way to Mercury. Progress was made on the joint EarthCARE Earth science mission.

The Director General and the Chairman of ISRO met virtually on 30 July and committed to re-energise bilateral cooperation between the two agencies. ESA and ISRO signed a Network and Cross-Support Arrangement on 21 June and worked together on a GNSS receiver calibration project.

The Director General and the Head of the Australian Space Agency held a virtual bilateral meeting on 29 April and jointly announced the construction of a second 35-metre, deep space antenna at ESA's New Norcia station.

ESA and CONAE continued their cooperation on the ESA deep space antenna in Argentina.

ESA and the Mexican Ministry of Foreign Affairs discussed possible future cooperation with the recently created Latin American and Caribbean Space Agency (ALCE).

Education

Except for a few in-person projects and experiment campaigns for university students, the Education Office continued to adapt to the ongoing pandemic while delivering all of its activities online.

In particular, it pursued:

- expansion of the ESERO network to include additional Members and Associate States;
- the enrichment of the ESA classroom resources portfolio to support teachers, children and parents with learning in continued home-schooling settings;
- the consolidation and improvement of the ESA school projects portfolio (Astro Pi, Climate Detectives, Moon Camp, CanSat, Mission X);
- the expansion of the ESA Academy Training and Learning Programme, through the development of new dedicated training sessions;
- the identification of launch opportunities for the educational CubeSat programme for university students;
- the development of a dedicated e-learning system to be systematically integrated, in the future, in the ESA Education offer.

The Education Office continued the development of Space for Education 2030; ESA's long-term Education Programme to meaningfully support the implementation of the five strategic priorities set out in ESA Agenda 2025, by attracting and retaining the best talents to the rapidly evolving space sector, for it to remain competitive; by supporting society in providing a long-term response to the

emerging global challenges (climate change, energy and resources management, security, health and safety) by means of education; by particularly addressing the green and digital transformations.

Corporate Development

Corporate Social Responsibility

ESA is a socially responsible organisation, taking maximum account of the needs of future generations regarding the sustainability of its activities vis-à-vis the environment, the economy and society at large.

ESA is committed, in collaboration and in coordination with its stakeholders, to pursuing the sustainability of its activities by:

1. increasing the contribution of space projects to the sustainable development of society;
2. improving the socially and environmentally responsible management of space sector activities.

After the appointment of the Chief Climate and Sustainability Officer in charge of ensuring full support for the implementation of the Paris Agreement and the European Green Deal by ESA and European space programmes, ESA has begun defining an Agency-wide sustainability agenda summarising the activities to be undertaken by the Agency to address social responsibility, environmental and sustainability issues for the benefit of society. This entails concrete actions to reduce the environmental footprint of its activities and the commitment to reduce its greenhouse gas (GHG) emissions by 46% by 2030, without neglecting actions on the preservation of biodiversity, human health and resource scarcity.

ESA has sought to marshal the efforts of the European space sector towards key sustainability topics by establishing joint initiatives with its public and private stakeholders and by organising a dedicated session of the Global Networking Forum on Challenges ahead for a Responsible Space Sector on 25 October during IAC2021, where the intention to collaborate was declared by heads of agencies and CEOs of Large System Integrators. The ESA Director General urged the joint action the Agency had been promoting for several months and all panellists expressed their commitment to working together towards common sustainability goals.

Advisory Bodies

ESA advisory groups can be divided into two categories: advisory groups to ESA delegate bodies and advisory groups to the ESA Director General.

Advisory Groups to ESA Delegate Bodies:

AFC Advisory Group

Pension Buffer Fund Management Board (PBF-MB)

IPC Advisory Groups

Technology Harmonisation Advisory Group (THAG)

Technology Advisory Working Group (TA-WG)

Industrial Policy Evolution Working Group (IPE-WG)

JCB Advisory Groups

5G Advisory Committee (5JAC)

Space Systems for Safety and Security (4S) Advisory Committee

Optical Communication – ScyLight Advisory Committee (SCOTT)

PB-EO Advisory Group

Data Operations Scientific and Technical Advisory Group (DOSTAG)

PB-HME Advisory Group

Exploration and Utilisation Board (EUB)

SEC Advisory Groups

ESA INFOSEC Panel

ESA Industrial Security Panel

Advisory groups to the ESA Director General:

Science Advisory Committees

Space Science Advisory Committee (SSAC)

Advisory Committee for Earth Observation (ACEO)

GNSS Science Advisory Committee (GSAC)

Human Spaceflight and Exploration Science Advisory Committee (HESAC)

Space Situational Awareness Advisory Group (SSA-AG)

Science Working Groups

Astronomy Working Group (AWG)

Life Sciences Working Group (LSWG)

Physical Sciences Working Group (PSWG)

Solar System Exploration Working Group (SSEWG)

Other Advisory Groups

Advisory Communication Committee (ACC)

Advisory Committee on Education (EXP-ACE).

More information on the various groups can be found in the yearly Council document on the ESA Advisory Structure.

Accounts

Statement of Financial Position

Statement of Financial Position

	Notes	31 December 2021 million €	31 December 2020 million €
Total Assets		15,859.4	14,135.5
Current assets		4,819.1	3,223.8
Cash and cash equivalents	3.1	1,954.4	1,573.4
Recoverables from non-exchange transactions	3.2	234.5	258.2
Receivables from exchange transactions	3.3	2,151.8	1,044.1
Inventories	3.4	478.4	348.1
Non-current assets		11,040.3	10,911.7
Recoverables from non-exchange transactions	3.2	12.0	3.7
Receivables from exchange transactions	3.3	666.4	1,086.9
Financial assets	3.5	1,164.2	959.0
Property, plant and equipment	3.6	8,874.6	8,557.6
Intangible assets	3.6	323.1	304.5
Total Liabilities		12,528.5	12,399.2
Current liabilities		3,248.6	2,865.1
Payables under exchange transactions		2,720.6	2,378.9
Current payables	3.7	1,516.7	1,321.2
Pre-financing from exchange transactions	3.8	1,072.7	976.5
Deferred income	3.9	131.2	81.2
Contributions and other payables		234.1	212.5
Prepaid contributions and other amounts payable to Member/Participating States	3.10	201.7	164.4
Pre-financing from non-exchange transactions		2.4	3.3
Other payables	3.11	30.0	44.8
Employee benefits obligations	3.12	232.7	221.7
Provisions	3.13	61.2	52.0
Non-current liabilities		9,279.9	9,534.1
Employee benefits obligations	3.12	8,749.3	9,420.8
Other payables	3.11	229.0	0.0
Deferred income	3.9	280.7	95.1
Pre-financing from non-exchange transactions		0.0	0.5
Pre-financing from exchange transactions	3.8	20.9	17.7
Net Assets		3,330.9	1,736.3
Reserves		6,352.4	4,152.3
Amounts to be called from Member States	3.14	-3,618.3	-3,375.0
Net Surplus/ (Deficit) for the period	II	586.4	959.0
Net Assets		3,330.9	1,736.3

Statement of Financial Performance

Statement of Financial Performance

	Notes	2021 million €	2020 million €
Operating Revenues	4.1	5,895.5	6,736.4
Contributions from Member/Participating States		4,534.5	4,800.2
Other contributions and grants		26.6	36.5
Revenues from exchange transactions		990.9	1,595.0
Staff contributions		55.3	49.1
Other operating income		288.2	255.6
Operating Expenses	4.1	-5,384.7	-5,853.3
Purchases		-2,146.8	-2,814.1
External services		-1,336.0	-1,086.5
Staff expenses		-745.5	-697.1
Other operating expenses		-273.6	-254.1
Expenses from non-exchange transactions (*)		-103.3	-165.2
Depreciation and amortization		-738.8	-803.0
Impairment		0.1	56.0
Write-downs inventories and fixed assets, doubtful debts, provisions for future losses		-40.8	-89.3
Net Surplus / (Deficit) from Operating activities		510.8	883.1
Financial activities	4.2	75.6	75.9
Financial revenue		99.8	116.0
Financial expenses		-24.2	-40.1
Net Surplus / (Deficit) for the period		586.4	959.0

(*) Expenses from non-exchange transactions are mainly related to the ESA/Arianespace Contract concerning the Ariane supplementary exploitation accompanying activities of the LEAP (the "LEAP Supplementary Frame Contract") (see also Note 3.11.)

Statement of Changes in Net Assets/Equity

Statement of Changes in Net Assets

		31 December 2019 restated	Allocation of audited 2019 surplus/ (deficit)	Movements 2020	31 December 2020
		million €	million €	million €	million €
RESERVES	A	4,371.0	333.8	-552.5	4,152.3
Member States Capital and Reserves		-3,158.5	-616.2	-499.1	-4,273.8
Revaluation Reserve		1,586.8	-501.4	-0.2	1,085.2
Currency Exchange Gains Reserve		2.9	0.2		3.1
Sundry Reserves		0.1			0.1
Other Reserves		-4,748.3	-115.0	-498.9	-5,362.1
<i>Social security reserve</i>		28.6	7.7	0.5	36.8
<i>Buffer Fund fair value reserve</i>		13.1		2.1	15.2
<i>Reserve Barter</i>		-283.1	-122.6		-405.7
<i>Accumulated actuarial gains and losses</i>		-4,506.9		-501.5	-5,008.5
Accumulated Surplus		7,529.4	950.0	-53.4	8,426.0
<i>Accumulated Surplus ESA programmes</i>		7,357.0	945.1	-0.8	8,301.3
<i>Accumulated Surplus Third Party programmes</i>		154.9	1.9	-52.6	104.2
<i>Accumulated Surplus PECS</i>		17.4	3.0		20.4
AMOUNTS TO BE CALLED FROM MEMBER STATES	B	-3,407.0	32.0	0.0	-3,375.0
Member States pension scheme receivable		-3,407.0	32.0		-3,375.0
Accumulated Deficit					
Net Surplus / (Deficit) for the Period	C	365.8	-365.8	959.0	959.0
TOTAL NET ASSETS	A+B+C	1,329.8	0.0	406.4	1,736.3

Statement of Changes in Net Assets

		31 December 2020	Allocation of audited 2020 surplus/ (deficit)	2021 Net Surplus	Change in the present value of the Employee Benefit Defined Obligation	Fair Value adjustments	Transfers between reserves	Other changes	31 December 2021
		million €	million €	million €	million €	million €	million €	million €	million €
RESERVES	A	4,152.3	1,202.2		1,017.8	-9.1	0.0	-0.4	6,362.8
Member States Capital and Reserves		-4,273.8	70.9		1,017.8	-9.1	-1.5		-3,195.5
Revaluation Reserve		1,085.2	-258.5				-1.5		825.2
Currency Exchange Gains Reserve		3.1	-1.5				0.1		1.7
Sundry Reserves		0.1							0.1
Other Reserves		-5,362.1	330.9		1,017.8	-9.1			-4,022.5
<i>Social security reserve</i>		36.8	9.1			-0.7			45.2
<i>Buffer Fund fair value reserve</i>		15.2				-8.4			6.9
<i>Reserve Barter</i>		-405.7	321.8						-83.9
<i>Actuarial gains and losses</i>		-5,008.5			1,017.8				-3,990.7
Accumulated Surplus		8,426.0	1,131.3				1.5	-0.4	9,558.3
<i>Accumulated Surplus ESA programmes</i>		8,301.3	1,104.7				1.5		9,407.5
<i>Accumulated Surplus Third Party programmes</i>		104.2	27.9					-0.4	131.7
<i>Accumulated Surplus PECS</i>		20.4	-1.3						19.1
AMOUNTS TO BE CALLED FROM MEMBER STATES	B	-3,375.0	-243.2	0.0	0.0	0.0	0.0	0.0	-3,618.3
Member States pension scheme receivable		-3,375.0	-243.2						-3,618.3
Accumulated Deficit									
Net Surplus / (Deficit) for the Period	C	959.0	-959.0	586.4					586.4
TOTAL NET ASSETS	A+B+C	1,736.3	0.0	586.4	1,017.8	-9.1	0.0	-0.4	3,330.9

Statement of Cash Flow

Statement of Cash Flows

	31 December 2021 million € 381.0	31 December 2020 million € 270.6
Net Cash Flows		
Net Cash Flows from Operating Activities	1,588.3	1,283.2
Surplus/(deficit) from ordinary activities	586.4	959.0
<i>Reclassification to investing activities</i>	-43.0	-73.3
<i>Non-cash movements</i>	732.0	862.9
Depreciation and impairment	738.8	747.0
Provisions and Accruals	34.7	-2.3
Write-downs / Adjustments inventories and fixed assets, provisions for future losses on inventories	-0.6	88.6
Revaluation of financial assets at fair value	-40.8	29.6
<i>Decrease (Increase) in Deferrals/accruals of past or future operating cash receipts or payments</i>	312.9	-465.4
Decrease/(increase) in inventories	-127.1	284.8
Decrease/(increase) in recoverables from Member States	23.7	154.9
Decrease/(increase) in other receivables	214.2	-709.0
Decrease/(increase) in other current assets	-935.3	-214.0
Increase/(decrease) in payable to Member/Participating States	37.3	-113.4
Increase/(decrease) in pre-financing	124.5	-68.9
Increase/(decrease) in other payables	618.4	-130.2
Increase/(decrease) in provisions for employee benefits	357.3	330.3
Net Cash Flows from Investing Activities	-1,207.3	-1,012.5
Acquisition of fixed assets	-1,076.9	-959.0
Purchases of financial investments (portfolio)	-341.2	-517.0
Proceeds on disposal of financial investments (portfolio)	205.2	455.7
Interests received on financial assets	5.6	7.8
Net increase/(decrease) in cash and cash equivalents	381.0	270.6
Cash and cash equivalents at the beginning of the year	1,573.4	1,302.7
Cash and cash equivalents at year-end	1,954.4	1,573.4

ANNEX

Annex 1 – Agreements signed in 2021

- The Agreement between the European Space Agency and the Deutsches Zentrum für Luft- und Raumfahrt e.V. (DLR) for ESA's assistance to DLR for Lidar Technology Development was signed in Paris, France, on 14 January 2021 by Johann-Dietrich Wörner, ESA Director General, and in Bonn, Germany, on 21 January 2021 by Walther Pelzer, DLR Board Member Space Administration and Head of the German Delegation to ESA. This Agreement entered into force on 21 January 2021 and shall remain in force until 31 December 2024, unless terminated earlier according to the provisions of Article 11 or extended by written agreement. (LEG/516)
- The Agreement between the European Space Agency and the Norwegian Space Agency concerning the provision of Technical Assistance and Expertise in the framework of the Commercial Space Transportation Services and Support to Member States Programme was signed in Paris, France, on 17 November 2020 by Daniel Neuenschwander, ESA Director of Space Transportation, and in Oslo, Norway, on 1 December 2020 by Christian Hauglie-Hanssen, Director General of the Norwegian Space Agency. Accordingly, this Agreement entered into force on 1 December 2020 for an initial period of five years and may tacitly be renewed on the same terms for successive periods of five years. (LEG/517)
- The Financial Framework Partnership Agreement between the European Commission, representing the European Union, the European Union Agency for the Space Programme and the European Space Agency on the implementation of the Union Space Programme and Horizon Europe was signed on 22 June 2021 in Brussels, Belgium, by the ESA Director General, Josef Aschbacher, and by Timo Pesonen, European Commission Director General, Directorate-General for Defence, Industry and Space, and by Rodrigo Da Costa, EUSPA Executive Director. Accordingly, this Agreement entered into force on 23 June 2021 and shall remain in force until 31 December 2027, unless terminated earlier pursuant to Article 70 (Grounds for termination). (LEG/518)
- The Contribution Agreement between the European Commission, representing the European Union, and the European Space Agency on the implementation of the Union Space Programme and Horizon Europe was signed in Brussels, Belgium on 22 June 2021 by the ESA Director General, Josef Aschbacher, and by Timo Pesonen, European Commission Director-General, Directorate-General for Defence, Industry and Space, Accordingly, this Agreement entered into force on 22 June 2021 and shall remain in force until 31 December 2032, subject to the provisions on early termination or suspension for the implementation stipulated in the Financial Framework Partnership Agreement. (LEG/519)
- The Contribution Agreement between the European Union Agency for the Space Programme and the European Space Agency on the implementation of the Union Space Programme was signed in Brussels, Belgium, on 22 June 2021 by the ESA Director General, Josef Aschbacher, and by Rodrigo da Costa, EUSPA Executive Director. Accordingly, this Agreement entered into

force on 22 June 2021 and shall remain in force until 31 December 2032, subject to the provisions on early termination or suspension of the implementation in the Financial Framework Partnership Agreement. (LEG/520)

- The Agreement between ESA and JAXA regarding the Martian Moons Exploration (MMX) Mission was signed on 4 February 2021 in Sagami-hara, Japan by Hitoshi Kuninaka, JAXA Vice President, and in Madrid, Spain, by Günther Hasinger, ESA Director of Science. Accordingly, this Agreement entered into force on 4 February 2021 for a period of 10 years, until 3 February 2031. (LEG/521)
- The Memorandum of Understanding between ESA and NASA concerning the Nancy Grace Roman Space Telescope Mission was signed on 13 July 2021 in Washington D.C., U.S., by Gilbert Kirkham, Science Division Director of NASA's Office of International and Interagency Relations, and on 22 July 2021 in Paris, France, by Frédéric Nordlund, Head of the ESA European and External Relations Department. Accordingly, this Memorandum of Understanding entered into force on 22 July 2021 and shall remain in force until five years plus 90 days after launch of the Roman spacecraft or until 30 June 2034, whichever is earlier. (LEG/522)
- The Association Agreement between the European Space Agency and the Government of the Republic of Lithuania was signed in Paris, France on 18 March 2021 by the ESA Director General, Josef Aschbacher, and in Vilnius, Lithuania, on 28 April 2021 by Aušrinė Armonaitė, Minister of Economy and Innovation of the Republic of Lithuania. Accordingly, the Agreement entered into force on 21 May 2021 for a period of seven years, until 21 May 2028. (LEG/523)
- The Agreement between ESA and JAXA regarding Cooperation on ESA's Hera Mission was signed on 4 February 2021 in Tokyo, Japan and Darmstadt, Germany, by the ESA Director General, Josef Aschbacher, and Yamakawa Hiroshi, President of JAXA. Accordingly, the Agreement entered into force on 4 February 2021 and shall remain in force until December 2027. (LEG/524)
- The Arrangement for ESA's Assistance to the Italian National Project concerning Earth Observation and Space Transportation was signed in Rome, Italy on 16 December 2021 in by the ESA Director General, Josef Aschbacher, and by Vittorio Colao, Minister for Technological Innovation and Digital Transition of Italy. Accordingly, the Arrangement entered into force on 16 December 2021 and shall remain in force until 31 December 2026. (LEG/525)

Annex 2 – Patent Groups

TABLE 1: LIST OF 2021 INVENTIONS THAT ESA DECIDED TO PROTECT

ESA ref	Title	Decision	Name of ESA inventors	Origin of the invention	Ownership
789	UPPER SIDE BAND SYMMETRIC SYNTONIZATION BETWEEN TWO SATELLITES	PCT patent application	M. Martin Neira	In-house work	ESA
780	Confidential- still to be protected	PCT patent application	F. Amarillo	In-housework	ESA
782	DEVELOPABLE WAVEGUIDE COUPLER FOR USE IN MICROWAVE DEVICES	PCT patent application	N. Fonseca	In-house work	ESA
783	“HEURISTIC RADIO RESOURCE MANAGEMENT FOR MASSIVE MIMO IN BROADBAND COMMUNICATION SATELLITES”	PCT patent application	R. de Gaudenzi & P. Angeletti	In-house work	ESA
792	TRANSMIT/RECEIVE MULTIPLE FEED PER BEAM SINGLE REFLECTOR ANTENNA	PCT patent application	N. Fonseca	In-house work	ESA

TABLE 2: LIST OF ESA PATENTS GRANTED BY PATENT OFFICES IN 2021

ESA/PAT	TITLE	COUNTRY OF GRANT
610	RECONFIGURABLE RF FRONT END CIRCUIT FOR A MULTI-BEAM ARRAY FED REFLECTOR ANTENNA SYSTEM	DE, FR, GB and CH
714	EXOTHERMIC REACTION AIDED SPACECRAFT DEMISE DURING RE-ENTRY	DE, FR, GB, AT and CH
653	HIGH VOLTAGE TOLERANT CMOS DRIVER FOR LOW-VOLTAGE BI-DIRECTIONAL COMMUNICATION BUSES	DE, FR, GB, ES and SE
735	MAGNETIC DAMPING FOR SPACECRAFT STABILISATION AFTER END OF LIFE	FR, GB, DE, PT, IT, ES, BE and US
675	INTERFERENCE-RESILIENT FLEXIBLE TECHNIQUES FOR PAYLOAD RESOURCE ALLOCATION IN BROADBAND SATELLITES	GB, LU, DE, ES, FR
654	A BUS FAILURE DETECTION, PROTECTION AND PROPAGATION PREVENTION CAN TRANSCEIVER ARCHITECTURE FOR LOW-VOLTAGE CMOS PROCESS IMPLEMENTATION	DE, NL, BE, CH, SE, ES, IT, AT, FR, and GB,
541	A METHOD FOR COMPENSATING SIGNAL DISTORTION IN AN EMITTING PAYLOAD PROCÉDÉ DE COMPENSATION DE DISTORSION DE SIGNAUX DANS UNE CHARGE UTILE ÉMETTRICE	IN
702	A DIRECTIONAL COUPLER AND METHOD OF MANUFACTURING THEREOF	US
720	COMPRESSION AND DECOMPRESSION OF FIXED LENGTH DATA STRUCTURES	US

Annex 3 – List of events organised in 2021

Date	Event
07.01.2021	Contract signature with Thales Alenia France for ESPRIT module for Gateway
14.01.2021	Director General's press conference
02.02.2021	Contract signature with Airbus for three more European Service Modules
01.03.2021	Josef Aschbacher becomes new ESA Director General
09.03.2021	Contract signature with OHB Sweden for a prototype Arctic Weather Satellite
23.04.2021	Launch of Thomas Pesquet's Alpha mission to the ISS
31.03.2021	Opening of ESA Astronaut Selection
07.04.2021	Director General presents ESA Agenda 2025 to the public
21.04.2021	20 years of Europeans on the ISS
29.04.2021	Vega launch SSMS adapter deploying 5 additional satellites
07.05.2021	Launch of Coldplay album in space with Thomas Pesquet on board the ISS
17.05.2021	Solar Orbiter's first images of coronal mass ejections
20.05.2021	Contract signatures for Moonlight with SSTL and Telespazio
21.05.2021	Lithuania becomes ESA Associate Member State
28.05.2021	Contract signatures for new generation of Galileo with Airbus and Thales Alenia
10.06.2021	Announcement of EnVision, the new ESA mission to Venus
22.06.2021	Signature of the Financial Framework Partnership Agreement with the European Commission
23.06.2021	First results of the Astronaut Selection
02.07.2021	First successful high-altitude drop tests for ExoMars parachute
21.07.2021	Launch of the European Robotic Arm to the ISS
30.07.2021	Launch of Eutelsat Quantum Telecommunication satellite

17.08.2021	Vega launch SSMS adapter deploying five additional satellites
03.09.2021	ESA Council agrees on resolution on Ariane 6 and Vega-C exploitation
28.09.2021	Ariane 6 launch complex inaugurated
02.10.2021	BepiColombo's first images of Mercury
03.10.2021	Virtual ESA Open Day at ESTEC, ECSAT, and ESRIN
04.10.2021	Release of ESA and Mattel's Samantha Cristoforetti Barbie doll
13.10.2021	Delivery of second European Service Module to NASA
20.10.2021	10 years of Soyuz launches from Kourou
21.10.2021	ESA Council appoints new ESA Directors
25.10.2021	ESA at the 72 nd International Astronautical Congress
27.10.2021	"ESA Champions" award initiative launched
09.11.2021	Thomas Pesquet returns from space after six months in orbit
11.11.2021	Launch of Matthias Maurer's Cosmic Kiss mission to the ISS
16.11.2021	Vega launch SSMS adapter deploying five additional satellites
19.11.2021	Intermediate Ministerial Meeting in Matosinhos, Portugal
05.12.2021	Launch of Galileo satellites 27-28
13.12.2021	Launch of ESA Playmobil Mars exploration set
25.12.2021	Launch of the James Webb Space Telescope

Annex 4 - List of ESA Publications

- ESA Agenda 2025: Executive Summary (EN/FR/DE/IT/ES; also interactive version)
- Webb (EN/FR/DE/IT/ES/NL; print and interactive)
- ESA Highlights magazine 2021 (print and interactive)
- ESA Highlights magazine 2020 (interactive)
- ESA Impact (interactive – 4x Quarterly and 2x Council)
- This is ESA (produced in Russian + Irish; reprinted in English + Italian)
- Cosmic Kiss mission brochure (in English and German)
- Alpha mission brochure (in English and French)